SET 21 • 7.NS.A.2.b

BACKWARD 2

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$\frac{4}{5} \div \frac{2}{15}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q1/5

802

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q1/5

Divide.

$3 \div \frac{3}{8}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q2/5

803

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q2/5

Divide.

$\frac{7}{10} \div \frac{1}{5}$

A. $\frac{2}{7}$

B. $\frac{7}{50}$

C. $1\frac{2}{5}$

D. $3\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q3/5

804

QUESTION ID

S21-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q3/5

True or false?

$\frac{3}{8} \div \frac{3}{4} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q4/5

805

QUESTION ID

S21-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

$\frac{5}{6} \div \frac{5}{12} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q5/5

806

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 7.NS.A.2.b

FORWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is $\frac{-5}{8}$ rational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q1/5

808

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. It is a quotient of the integers −5 and 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q1/5

Which number is irrational?

A. $\sqrt{7}$

B. 0.25

C. $\frac{-22}{7}$

D. $-\sqrt{16}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q2/5

809

QUESTION ID

S21-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. √7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q2/5

Write −2.4 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q3/5

810

QUESTION ID

S21-F1-Q3

TYPE

Short answer

CORRECT ANSWER

−12/5

GRADING NOTE Any equivalent quotient, such as −24/10, is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q3/5

True or false?

Every integer is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q4/5

811

QUESTION ID

S21-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q4/5

True or false?

√25 is irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q5/5

812

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 7.NS.A.2.b

FORWARD 2

Decide whether a product is rational or irrational • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3

Is $-3 \times \sqrt{5}$ rational or irrational?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q1/5

814

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Irrational

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q1**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q1/5

Is $\frac{-2}{3} \times \frac{9}{4}$ rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q2/5

815

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Rational; −3/2

GRADING NOTE −1 1/2 or −1.5 is also correct for the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q2/5

Which product is rational?

A. $-1 \times \pi$

B. $5 \times \sqrt{2}$

C. $2 \times \sqrt{3}$

D. $\sqrt{2} \times \sqrt{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q3/5

816

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\sqrt{2} \times \sqrt{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q3/5

True or false?

The product of a nonzero rational number and an irrational number is irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q4/5

817

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q4/5

True or false?

The quotient of two rational numbers (with a nonzero divisor) can be irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q5/5

818

QUESTION ID

S21-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q5/5

SET 22 • 7.NS.A.2.b • THE NUMBER SYSTEM

Interpret quotients of rational numbers in real-world situations

MAIN	Grade 7 main question	**7.NS.A.2.b**
BACKWARD 1	Divide fractions in word problems	**6.NS.A.1**
BACKWARD 2	Divide unit fractions and whole numbers in word problems	**5.NF.B.7.c**
FORWARD 1	Find a rate of change from a description	**8.F.B.4**
FORWARD 2	Calculate and interpret a negative average rate of change	**HSF-IF.B.6**

SET 22 • 7.NS.A.2.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.b

A snail's position changed by $-\frac{3}{4}$ meter in $\frac{1}{2}$ minute at a steady rate.

What was the change in position per minute?

Change per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q1/5

821

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

−3/2 meters per minute

GRADING NOTE −1 1/2 or −1.5 meters per minute is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q1/5

A diver's position changed by −45 feet in 9 seconds at a steady rate.

What was the change in position per second?

Change per second: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q2/5

822

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

−5 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q2/5

The temperature changed by $-4\frac{1}{2}$°F over $1\frac{1}{2}$ hours at a steady rate.

What was the change per hour?

A. $-\frac{1}{3}$°F

B. 3°F

C. −3°F

D. $-6\frac{3}{4}$°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q3/5

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −3°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q3/5

True or false?

A balance changes by −$120 over 8 weeks at a steady rate. The change is −$15 per week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q4/5

824

QUESTION ID

S22-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q4/5

True or false?

A water level changed by $-2\frac{2}{5}$ inches over 4 days at a steady rate. The change was $\frac{3}{5}$ inch per day.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q5/5

825

QUESTION ID

S22-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The change was −3/5 inch per day.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q5/5

SET 22 • 7.NS.A.2.b

BACKWARD 1

Divide fractions in word problems • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{3}{4}$-cup servings are in 6 cups of rice?

Servings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q1/5

827

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q1/5

A $\frac{2}{3}$-mile trail is split into 4 equal parts.

How long is each part?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q2/5

828

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/6 mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q2/5

A strip $\frac{5}{6}$ meter long is cut into pieces $\frac{1}{12}$ meter long.

How many pieces are there?

A. $\frac{5}{72}$

B. 12

C. 6

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q3/5

829

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q3/5

True or false?

There are 6 half-pound portions in 3 pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q4/5

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q4/5

True or false?

3 people share $\frac{3}{4}$ pound of nuts equally. Each gets $\frac{1}{2}$ pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q5/5

831

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each gets 1/4 pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 7.NS.A.2.b

BACKWARD 2

Divide unit fractions and whole numbers in word problems • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c

How many $\frac{1}{4}$-pound burgers can be made from 5 pounds of meat?

Burgers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q1/5

833

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q1/5

3 friends share $\frac{1}{2}$ pound of grapes equally.

How much does each friend get?

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q2/5

834

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/6 pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q2/5

2 people share $\frac{1}{3}$ of a pizza equally.

How much of the whole pizza does each person get?

A. $\frac{2}{3}$ of the pizza

B. $\frac{1}{6}$ of the pizza

C. 6 pizzas

D. $\frac{3}{2}$ of the pizza

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q3/5

835

QUESTION ID

S22-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/6 of the pizza

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q3/5

True or false?

4 pounds of rice is packed into bags that each hold $\frac{1}{5}$ pound. There are 20 bags.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q4/5

836

QUESTION ID

S22-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q4/5

True or false?

2 people share $\frac{1}{8}$ pound of nuts equally. Each person gets $\frac{1}{4}$ pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q5/5

837

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each person gets 1/16 pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 7.NS.A.2.b

FORWARD 1

Find a rate of change from a description • Grade 8 • Functions • 8.F.B.4

A tank held 240 gallons at 2:00 and 180 gallons at 2:15. It drained at a constant rate.

What is the rate of change in gallons per minute?

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q1/5

839

QUESTION ID

S22-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−4 gallons per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q1/5

A plane descends from 30,000 feet to 21,000 feet in 15 minutes at a constant rate.

What is the rate of change in feet per minute?

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q2/5

840

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−600 feet per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q2/5

A line passes through (0, 50) and (10, 20).

What is its rate of change?

A. $-\frac{1}{3}$

B. 3

C. -30

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q3/5

841

QUESTION ID

S22-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q3/5

True or false?

A balance that goes from $80 to $20 in 12 weeks at a constant rate changes by −$5 per week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q4/5

842

QUESTION ID

S22-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q4/5

True or false?

A temperature that goes from 10° to −6° in 4 hours at a constant rate changes by −1° per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q5/5

843

QUESTION ID

S22-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It changes by −4° per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 7.NS.A.2.b

FORWARD 2

Calculate and interpret a negative average rate of change • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

The table shows temperatures.

What is the average rate of change from 0 hours to 6 hours?

Time (h)	0	3	6
Temperature (°F)	12	3	−6

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q1/5

845

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−3°F per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q1/5

$f(x) = -x^2 + 10$

What is the average rate of change from $x = 1$ to $x = 3$?

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q2/5

846

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q2/5

A company's profit was $5,000 in year 2 and $2,000 in year 5.

What is the average rate of change?

A. −$600 per year

B. $1,000 per year

C. −$3,000 per year

D. −$1,000 per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q3/5

847

QUESTION ID

S22-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −$1,000 per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q3/5

True or false?

A negative average rate of change means the quantity decreased over the interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q4/5

848

QUESTION ID

S22-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q4/5

True or false?

For f(x) = 5 − 2x, the average rate of change from x = 0 to x = 4 is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q5/5

849

QUESTION ID

S22-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q5/5

SET 23 • 7.NS.A.2.c • THE NUMBER SYSTEM

Use properties of operations to multiply and divide rational numbers

MAIN	Grade 7 main question	**7.NS.A.2.c**
BACKWARD 1	Apply the distributive property	**6.EE.A.3**
BACKWARD 2	Use the commutative and associative properties of multiplication	**3.OA.B.5**
FORWARD 1	Solve equations with fraction coefficients on both sides	**8.EE.C.7.b**
FORWARD 2	Rewrite expressions with rational exponents	**HSN-RN.A.2**

SET 23 • 7.NS.A.2.c

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.c

Use properties to multiply. Show how you grouped or ordered the factors.

$-4 \times 7.3 \times 25$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q1/5

852

QUESTION ID

S23-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

−730

GRADING NOTE REQUIRED WORK: Any correct use of the commutative and associative properties. Example: $(-4 \times 25) \times 7.3 = -100 \times 7.3$. Other valid groupings are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q1/5

Rewrite the division as multiplication by a reciprocal. Then use properties to find the value. Show your work.

$-18 \div \frac{3}{4} \times \frac{1}{6}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q2/5

853

QUESTION ID

S23-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

−4

GRADING NOTE REQUIRED WORK: Division rewritten as multiplying by $\frac{4}{3}$, then any valid regrouping. Example: $-18 \times \frac{4}{3} \times \frac{1}{6} = (-18 \times \frac{1}{6}) \times \frac{4}{3} = -3 \times \frac{4}{3}$. Other valid groupings are accepted; division itself is not reordered.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q2/5

Which property is shown?

$-2 \times (5 \times 3.1) = (-2 \times 5) \times 3.1$

A. Multiplicative identity property

B. Associative property of multiplication

C. Distributive property

D. Commutative property of multiplication

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q3/5

854

QUESTION ID

S23-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Associative property of multiplication

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q3/5

True or false?

$-6 \times (10 + \frac{1}{2}) = -60 + (-3)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q4/5

855

QUESTION ID

S23-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q4/5

True or false?

$-24 \div 6 \div 2 = -24 \div (6 \div 2)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q5/5

856

QUESTION ID

S23-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division is not associative: $-24 \div 6 \div 2 = -2$, but $-24 \div 3 = -8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q5/5

SET 23 • 7.NS.A.2.c

BACKWARD 1

Apply the distributive property • Grade 6 • Expressions & Equations • 6.EE.A.3

Rewrite using the distributive property.

7(2y + 3)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q1/5

858

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

14y + 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the distributive property to find 8 × 47 as 8 × (40 + 7).

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q2/5

859

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

376

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q2/5

Which expression is equivalent to 5(x + 9)?

A. 5x + 14

B. x + 45

C. 5x + 45

D. 5x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q3/5

860

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5x + 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q3/5

True or false?

$4(3 + 2.5) = 12 + 10$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q4/5

861

QUESTION ID

S23-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q4/5

True or false?

$3(a + 4) = 3a + 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q5/5

862

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 7.NS.A.2.c

BACKWARD 2

Use the commutative and associative properties of multiplication • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

Fill in the blank.

$6 \times 4 = 4 \times ___$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

864

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

Find 2 × 7 × 5 by first multiplying 2 × 5.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q2/5

865

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q2/5

Which expression is equal to (3 × 2) × 4?

A. 3 × 2 + 4

B. 3 + (2 × 4)

C. 3 × (2 × 4)

D. (3 + 2) × 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

866

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3 × (2 × 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

True or false?

$8 \times 3 = 3 \times 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q4/5

867

QUESTION ID

S23-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q4/5

True or false?

$(2 \times 3) \times 5 = 2 \times 3 + 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q5/5

868

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 7.NS.A.2.c

FORWARD 1

Solve equations with fraction coefficients on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$\frac{3}{4}x - 2 = \frac{1}{2}x + 3$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q1/5

870

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q1/5

Solve for x.

$-\frac{2}{3}(x + 6) = \frac{1}{3}x - 1$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q2/5

871

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q2/5

Solve for x.

$\frac{x}{4} + \frac{x}{6} = 5$

A. x = 12

B. x = 6

C. x = 50

D. x = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q3/5

872

QUESTION ID

S23-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q3/5

True or false?

$x = -10$ is the solution of $-\frac{1}{2}x + 1 = \frac{1}{5}x + 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q4/5

873

QUESTION ID

S23-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q4/5

True or false?

$x = 3$ is the solution of $\frac{5}{6}x - 1 = \frac{1}{2}x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q5/5

874

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q5/5

SET 23 • 7.NS.A.2.c

FORWARD 2

Rewrite expressions with rational exponents • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

Find the value.

27^(1/3)

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q1/5

876

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q1/5

For a real number $x \geq 0$, write $\sqrt{x}$ using a rational exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q2/5

877

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x^(1/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q2/5

Which number is equal to 16^(1/2)?

A. 4

B. 32

C. 2

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q3/5

878

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q3/5

True or false?

$(x^3)^{(1/3)} = x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q4/5

879

QUESTION ID

S23-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

25^(1/2) = 12.5

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q5/5

880

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 25^(1/2) = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q5/5

SET 24 • 7.NS.A.2.d • THE NUMBER SYSTEM

Use long division to write a rational number as a terminating or repeating decimal

MAIN	Grade 7 main question	**7.NS.A.2.d**
BACKWARD 1	Write fractions with denominators of 10 or 100 as decimals	**4.NF.C.6**
BACKWARD 2	Divide multi-digit whole numbers	**6.NS.B.2**
FORWARD 1	Write a repeating decimal as a fraction	**8.NS.A.1**
FORWARD 2	Report a decimal result to a sensible level of accuracy	**HSN-Q.A.3**

SET 24 • 7.NS.A.2.d

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.d

ANSWER KEY Reduced copy of the question page, with its answer

Use long division to write $\frac{3}{8}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q1/5

883

QUESTION ID

S24-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

0.375

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q1/5

Use long division to write $\frac{5}{11}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q2/5

884

QUESTION ID

S24-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

0.4545... (the digits 45 repeat)

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

A bar over 45 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q2/5

Which decimal is equal to $-\frac{7}{12}$?

A. −0.58333... (the 3 repeats)

B. −0.712

C. −1.714...

D. −0.5833 (the decimal ends)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q3/5

885

QUESTION ID

S24-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −0.58333... (the 3 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q3/5

True or false?

$\frac{2}{9} = 0.222...$, a repeating decimal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q4/5

886

QUESTION ID

S24-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q4/5

True or false?

$\frac{7}{20}$ is a repeating decimal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q5/5

887

QUESTION ID

S24-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 7/20 = 0.35, a terminating decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q5/5

SET 24 • 7.NS.A.2.d

BACKWARD 1

Write fractions with denominators of 10 or 100 as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{7}{10}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q1/5

889

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q1/5

Write $\frac{43}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q2/5

890

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.43

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{6}{100}$?

A. 0.06

B. 0.006

C. 0.6

D. 6.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q3/5

891

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q3/5

True or false?

$\frac{9}{10} = 0.9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q4/5

892

QUESTION ID

S24-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q4/5

True or false?

$\frac{25}{100} = 2.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q5/5

893

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 7.NS.A.2.d

BACKWARD 2

Divide multi-digit whole numbers • Grade 6 • The Number System • 6.NS.B.2

ANSWER KEY Reduced copy of the question page, with its answer

Divide.

7,425 ÷ 25

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q1/5

895

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

297

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q1/5

Divide.

4,368 ÷ 52

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q2/5

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q2/5

Divide.

9,072 ÷ 36

A. 2,520

B. 252

C. 25.2

D. 242

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q3/5

897

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 252

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

6,912 ÷ 48 = 144

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q4/5

898

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

2,925 ÷ 75 = 49

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q5/5

899

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 7.NS.A.2.d

FORWARD 1

Write a repeating decimal as a fraction • Grade 8 • The Number System • 8.NS.A.1

Write 0.777... (the 7 repeats) as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q1/5

901

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q1/5

Write 0.1818... (the digits 18 repeat) as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q2/5

902

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2/11

GRADING NOTE 18/99 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q2/5

Which fraction is equal to 0.333... (the 3 repeats)?

A. $\frac{1}{3}$

B. $\frac{3}{100}$

C. $\frac{3}{10}$

D. $\frac{33}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q3/5

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q3/5

True or false?

0.999... (the 9 repeats) is equal to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q4/5

QUESTION ID

S24-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q4/5

True or false?

0.4545... (the digits 45 repeat) is equal to $\frac{45}{100}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q5/5

905

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is equal to 45/99 = 5/11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 7.NS.A.2.d

FORWARD 2

Report a decimal result to a sensible level of accuracy • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

A 10-foot board is cut into 3 equal pieces. The tape measure reads to the nearest hundredth of a foot.

Report the length of each piece to that accuracy.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q1/5

907

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3.33 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q1/5

$\frac{2}{7}$ = 0.285714... Round it to the nearest thousandth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q2/5

908

QUESTION ID

S24-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.286

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q2/5

A scale reads to the nearest 0.1 kg. 5 kg of flour is split into 3 equal bags.

Which report of each bag's mass fits the scale?

A. 1.666666 kg

B. 1.7 kg

C. 2 kg

D. 1.67 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q3/5

909

QUESTION ID

S24-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.7 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q3/5

True or false?

Reporting $\frac{1}{3}$ hour as 0.33 hour is accurate to the nearest hundredth of an hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q4/5

910

QUESTION ID

S24-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q4/5

True or false?

A length reported as 4.0 cm is less precise than a length reported as 4 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q5/5

911

QUESTION ID

S24-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q5/5

SET 25 • 7.NS.A.3 • THE NUMBER SYSTEM

Solve real-world problems with the four operations on rational numbers

MAIN	Grade 7 main question	**7.NS.A.3**
BACKWARD 1	Multiply and divide decimals in word problems	**6.NS.B.3**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Solve real-world problems with scientific notation	**8.EE.A.4**
FORWARD 2	Use units to solve multistep problems	**HSN-Q.A.1**

SET 25 • 7.NS.A.3

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.3

A freezer is at 4°C. It cools 1.5°C every hour.

What is the temperature after 6 hours?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q1/5

914

QUESTION ID

S25-M-Q1

TYPE

Short answer

CORRECT ANSWER

−5°C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q1/5

Maya owes $42.60. She pays it off in 6 equal payments.

How much is each payment?

Payment: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q2/5

915

QUESTION ID

S25-M-Q2

TYPE

Short answer

CORRECT ANSWER

$7.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q2/5

A diver is at −18 m. She rises $\frac{1}{3}$ of the distance to the surface.

What is her new position?

A. −6 m

B. 12 m

C. −12 m

D. −24 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q3/5

916

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −12 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q3/5

True or false?

A hiker is at −86 m. She climbs 120 m. She ends 34 m above sea level.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q4/5

917

QUESTION ID

S25-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q4/5

True or false?

Three stock prices change by −$2.50, +$4, and −$3.25. The total change is +$1.75.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q5/5

QUESTION ID

S25-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The total change is −$1.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q5/5

SET 25 • 7.NS.A.3

BACKWARD 1

Multiply and divide decimals in word problems • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Gas costs $3.85 per gallon.

How much do 12 gallons cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q1/5

920

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$46.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q1/5

A 7.5-meter rope is cut into 3 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q2/5

921

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.5 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q2/5

A 4.8-liter jug fills cups that hold 0.3 liter each.

How many cups can it fill?

A. 14.4

B. 16

C. 160

D. 1.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q3/5

922

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q3/5

True or false?

A 2.5-pound bag of nuts costs $3.20 per pound. The bag costs $8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q4/5

923

QUESTION ID

S25-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q4/5

True or false?

$9.60 shared equally by 4 people is $2.60 each.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q5/5

924

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each gets $2.40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 7.NS.A.3

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

x + 4.5 = 12

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q1/5

926

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for y.

6y = 39

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q2/5

927

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q2/5

Solve for n.

$\frac{2}{3}n = 10$

A. $n = 30$

B. $n = \frac{20}{3}$

C. $n = 15$

D. $n = 9\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q3/5

928

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. n = 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q3/5

True or false?

m = 8 is the solution of m − 2.75 = 5.25.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q4/5

929

QUESTION ID

S25-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q4/5

True or false?

k = 3 is the solution of 4k = 1.2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q5/5

930

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is k = 0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 7.NS.A.3

FORWARD 1

Solve real-world problems with scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A hard drive holds 2×10^{12} bytes. A file is 4×10^{8} bytes.

How many files of this size fit on the drive?

Files: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q1/5

932

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5,000

GRADING NOTE 5×10^{3} is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q1/5

Earth is about 1.5×10^8 km from the Sun. Mars is about 2.3×10^8 km from the Sun.

About how much farther from the Sun is Mars? Write the answer in scientific notation.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q2/5

933

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

8×10^7 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q2/5

A cell is 2×10^{-5} m wide.

How wide are 3,000 of these cells placed side by side?

A. 6×10^{-8} m

B. 5×10^{-2} m

C. 6×10^{-2} m

D. 6×10^{2} m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q3/5

934

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6×10^{-2} m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q3/5

True or false?

$(6 \times 10^4) \times 5 = 3 \times 10^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q4/5

935

QUESTION ID

S25-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q4/5

True or false?

$(9 \times 10^6) \div (3 \times 10^2) = 3 \times 10^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q5/5

936

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The quotient is 3×10^4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q5/5

SET 25 • 7.NS.A.3

FORWARD 2

Use units to solve multistep problems • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

A car gets 32 miles per gallon. Gas costs $3.60 per gallon.

Use units to find the gas cost per mile. Then find the cost of a 400-mile trip.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q1/5

938

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$0.1125 per mile; $45 for 400 miles

GRADING NOTE Grade both: $0.1125 (about 11.25 cents) per mile, and $45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q1/5

A faucet leaks 12 milliliters per minute.

How many liters leak in 1 day? Show the unit conversions.

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q2/5

939

QUESTION ID

S25-F2-Q2

TYPE

Short answer

CORRECT ANSWER

17.28 liters

GRADING NOTE 12 mL/min × 60 min/h × 24 h/day = 17,280 mL = 17.28 L.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q2/5

A printer uses 0.8 mL of ink per page. A 30 mL ink cartridge costs $24.

What is the cost of ink per page?

A. $0.64

B. $1.25

C. $0.32

D. $0.80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q3/5

940

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $0.64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q3/5

True or false?

A speed of 12 meters per second is 43.2 kilometers per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q4/5

941

QUESTION ID

S25-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q4/5

True or false?

Walking 3 miles per hour for 45 minutes covers 2.5 miles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q5/5

942

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It covers 2.25 miles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q5/5

SET 26 • 7.EE.A.1 • EXPRESSIONS & EQUATIONS

Add and subtract linear expressions with rational coefficients

MAIN	Grade 7 main question	**7.EE.A.1**
BACKWARD 1	Combine like terms	**6.EE.A.3**
BACKWARD 2	Identify equivalent expressions	**6.EE.A.4**
FORWARD 1	Solve equations by expanding and collecting like terms	**8.EE.C.7.b**
FORWARD 2	Interpret the terms, factors, and coefficients of an expression in a…	**HSA-SSE.A.1.a**

SET 26 • 7.EE.A.1

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$(3x + 5) + (2x - 8)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q1/5

945

QUESTION ID

S26-M-Q1

TYPE

Short answer

CORRECT ANSWER

5x − 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q1/5

Simplify.

$(4.5y - 2) - (1.5y + 6)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q2/5

946

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

3y − 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q2/5

Simplify.

$\frac{1}{2}a + 3 - \frac{3}{4}a + 1$

A. $-\frac{1}{4}a + 4$

B. $-\frac{1}{4}a + 2$

C. $\frac{1}{4}a + 4$

D. $\frac{5}{4}a + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q3/5

947

QUESTION ID

S26-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −(1/4)a + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q3/5

True or false?

$(6m - 1) - (2m - 4) = 4m + 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q4/5

948

QUESTION ID

S26-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q4/5

True or false?

$-3k + 7 + 5k - 2 = 8k + 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q5/5

949

QUESTION ID

S26-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The expression simplifies to 2k + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q5/5

SET 26 • 7.EE.A.1

BACKWARD 1

Combine like terms • Grade 6 • Expressions & Equations • 6.EE.A.3

Combine like terms.

5n + 3n + 4

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q1/5

951

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8n + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q1/5

Combine like terms.

2y + 7 + y

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q2/5

952

QUESTION ID

S26-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3y + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q2/5

Which expression is equivalent to x + x + x + 6?

A. 9x

B. 3x + 2

C. 3x + 6

D. $x^3 + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q3/5

953

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3x + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q3/5

True or false?

4b + 2b = 6b

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q4/5

954

QUESTION ID

S26-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q4/5

True or false?

$3a + 2 = 5a$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q5/5

955

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 7.EE.A.1

BACKWARD 2

Identify equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.4

Are 2(x + 3) and 2x + 6 equivalent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q1/5

957

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Using the distributive property, 2(x + 3) = 2x + 6 for every value of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q1/5

Which expression is equivalent to y + y + y?

A. 3y

B. y + 3

C. y^3

D. 3 + y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q2/5

958

QUESTION ID

S26-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 3y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q2/5

Write an expression with no parentheses that is equivalent to 4(n + 1).

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q3/5

959

QUESTION ID

S26-B2-Q3

TYPE

Short answer

CORRECT ANSWER

4n + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q3/5

True or false?

5m − 2m and 3m are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q4/5

960

QUESTION ID

S26-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q4/5

True or false?

a + 4 and 4a are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q5/5

961

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 7.EE.A.1

FORWARD 1

Solve equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$2(3x - 1) - (x + 4) = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q1/5

963

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q1/5

Solve for x.

$\frac{1}{2}(4x + 6) = 3x - 5$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q2/5

964

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q2/5

Solve for x.

$0.5(x + 8) - 2x = 1$

A. $x = 6$

B. $x = 3$

C. $x = 2$

D. $x = -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q3/5

965

QUESTION ID

S26-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q3/5

True or false?

$x = 5$ is the solution of $3(x - 2) = 2x - 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q4/5

966

QUESTION ID

S26-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q4/5

True or false?

$x = 1$ is the solution of $4x - (x + 2) = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q5/5

967

QUESTION ID

S26-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q5/5

SET 26 • 7.EE.A.1

FORWARD 2

Interpret the terms, factors, and coefficients of an expression in context • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A phone plan costs 0.1m + 25 dollars for m minutes in a month.

What do the terms 0.1m and 25 represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q1/5

969

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.1m is the cost of the minutes at $0.10 per minute; 25 is a fixed monthly charge of $25.

GRADING NOTE Grade both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F2-Q1**

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q1/5

The perimeter of a rectangle is 2(l + 4) centimeters, where l is its length.

What does the factor (l + 4) represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q2/5

970

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The length plus the width (the width is 4 cm), which is half of the perimeter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q2/5

A ride costs 2.5 + 1.75d dollars for d miles.

What does 1.75 represent?

A. The starting fee

B. The cost per mile

C. The total cost of a 1-mile ride

D. The number of miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q3/5

971

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The cost per mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q3/5

True or false?

A repair costs 40 + 15h dollars for h hours of work. The term 15h is the labor cost for h hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q4/5

QUESTION ID

S26-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q4/5

True or false?

A savings account holds 500 − 20w dollars after w weeks. The coefficient −20 means $20 is added each week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q5/5

973

QUESTION ID
S26-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE −20 means $20 is taken out each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q5/5

SET 27 • 7.EE.A.1 • EXPRESSIONS & EQUATIONS

Factor and expand linear expressions with rational coefficients

MAIN	Grade 7 main question	**7.EE.A.1**
BACKWARD 1	Use the greatest common factor and the distributive property	**6.NS.B.4**
BACKWARD 2	Identify the parts of an expression	**6.EE.A.2.b**
FORWARD 1	Expand to decide how many solutions an equation has	**8.EE.C.7.a**
FORWARD 2	Factor quadratic expressions	**HSA-SSE.A.2**

SET 27 • 7.EE.A.1

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$-4(2x - 3)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q1/5

976

QUESTION ID

S27-M-Q1

TYPE

Short answer

CORRECT ANSWER

$-8x + 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor out the greatest common factor.

6y + 15

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q2/5

977

QUESTION ID

S27-M-Q2

TYPE

Short answer

CORRECT ANSWER

3(2y + 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q2/5

Which expression is equivalent to $\frac{1}{2}(8a - 6)$?

A. $8a - 3$

B. $4a - 3$

C. $4a - 6$

D. $16a - 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q3/5

978

QUESTION ID

S27-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4a − 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q3/5

True or false?

$12m - 18 = 6(2m - 3)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q4/5

979

QUESTION ID

S27-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$-5(n + 2) = -5n + 10$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q5/5

980

QUESTION ID

S27-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-5(n + 2) = -5n - 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q5/5

SET 27 • 7.EE.A.1

BACKWARD 1

Use the greatest common factor and the distributive property • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q1/5

982

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q1/5

Use the greatest common factor to write 30 + 42 as a product.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q2/5

983

QUESTION ID

S27-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6(5 + 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q2/5

Which expression is equal to 18 + 27?

A. 9(2 + 3)

B. 9(2 + 27)

C. 9(18 + 3)

D. 3(6 + 27)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q3/5

QUESTION ID

S27-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9(2 + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q3/5

True or false?

The greatest common factor of 16 and 40 is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q4/5

985

QUESTION ID

S27-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q4/5

True or false?

$20 + 35 = 5(4 + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q5/5

986

QUESTION ID

S27-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $20 + 35 = 5(4 + 7)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 7.EE.A.1

BACKWARD 2

Identify the parts of an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.b

How many terms are in 3x + 7y − 2?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q1/5

988

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q1/5

What is the coefficient of w in 9 + 5w?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q2/5

989

QUESTION ID

S27-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which part of 4(n + 6) is a sum?

A. 4n

B. 4

C. n + 6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q3/5

990

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. n + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q3/5

True or false?

In 2(a + b), the factor 2 multiplies the sum a + b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q4/5

991

QUESTION ID

S27-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q4/5

True or false?

In 8k − 3, the term 8k is a constant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q5/5

992

QUESTION ID

S27-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q5/5

SET 27 • 7.EE.A.1

FORWARD 1

Expand to decide how many solutions an equation has • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$4(x - 2) = 4x - 8$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q1/5

994

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q1/5

How many solutions does this equation have?

$-3(x + 1) = -3x + 3$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q2/5

995

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q2/5

Which equation has no solution?

A. 2(x + 5) = x + 10

B. 2(x + 5) = 3x

C. 2(x + 5) = 2x + 10

D. 2(x + 5) = 2x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q3/5

996

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2(x + 5) = 2x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q3/5

True or false?

$6(x - 1) = 2(3x - 3)$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q4/5

997

QUESTION ID

S27-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q4/5

True or false?

5(x + 1) = 5x + 1 has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q5/5

998

QUESTION ID

S27-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q5/5

SET 27 • 7.EE.A.1

FORWARD 2

Factor quadratic expressions • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor.

$x^2 + 7x + 12$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q1/5

1000

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 3)(x + 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor completely.

$3x^2 - 12$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q2/5

1001

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3(x + 2)(x − 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q2/5

Factor.

$x^2 - 2x - 15$

A. $(x - 5)(x - 3)$

B. $(x - 15)(x + 1)$

C. $(x + 5)(x - 3)$

D. $(x - 5)(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q3/5

1002

QUESTION ID

S27-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $(x - 5)(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q3/5

True or false?

$2x^2 + 10x = 2x(x + 5)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q4/5

1003

QUESTION ID

S27-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 + 5x + 6 = (x + 1)(x + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q5/5

1004

QUESTION ID

S27-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 + 5x + 6 = (x + 2)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q5/5

SET 28 • 7.EE.A.2 • EXPRESSIONS & EQUATIONS

Rewrite an expression to show how the quantities in a problem are related

MAIN	Grade 7 main question	**7.EE.A.2**
BACKWARD 1	Write expressions from words	**6.EE.A.2.a**
BACKWARD 2	Write and interpret numerical expressions	**5.OA.A.2**
FORWARD 1	Write equivalent expressions with the properties of integer expon…	**8.EE.A.1**
FORWARD 2	Rewrite percent-growth expressions to reveal the growth factor	**HSA-SSE.B.3**

SET 28 • 7.EE.A.2

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.A.2

A shirt costs p dollars. The price is raised by 30%.

Write the new price as a single term.

New price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q1/5

1007

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

1.3p

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle's length is 4 cm more than its width w.

Write its perimeter in two equivalent ways.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q2/5

1008

QUESTION ID

S28-M-Q2

TYPE

Short answer

CORRECT ANSWER

2w + 2(w + 4) and 4w + 8

GRADING NOTE Any two correct equivalent expressions earn credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q2/5

A price p is lowered by 15%.

Which expression shows the new price?

A. 1.15p

B. 0.85p

C. 0.15p

D. p − 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q3/5

1009

QUESTION ID

S28-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.85p

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q3/5

True or false?

n + 0.08n = 1.08n shows that increasing n by 8% is the same as multiplying n by 1.08.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q4/5

1010

QUESTION ID

S28-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q4/5

True or false?

The price after a 20% discount on a price of d dollars is d − 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q5/5

1011

QUESTION ID

S28-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The price is 0.8d.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q5/5

SET 28 • 7.EE.A.2

BACKWARD 1

Write expressions from words • Grade 6 • Expressions & Equations • 6.EE.A.2.a

ANSWER KEY Reduced copy of the question page, with its answer

Write an expression for "5 less than y."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q1/5

1013

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

y − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an expression for "the product of 8 and n."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q2/5

1014

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q2/5

Which expression means "the sum of x and 3, times 4"?

A. 4(x + 3)

B. x + 3 × 4

C. 4x + 3

D. 4 + x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q3/5

1015

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q3/5

True or false?

"Twice a number m" can be written as 2m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q4/5

1016

QUESTION ID

S28-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q4/5

True or false?

"7 fewer than k" can be written as 7 − k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q5/5

1017

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is k − 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 7.EE.A.2

BACKWARD 2

Write and interpret numerical expressions • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write an expression for "add 6 and 9, then multiply by 2."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q1/5

1019

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2 × (6 + 9)

GRADING NOTE (6 + 9) × 2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q1/5

How does 5 × (24 + 13) compare with 24 + 13?

A. It is 5 more.

B. It is 5 less.

C. It is 5 times as large.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q2/5

1020

QUESTION ID

S28-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. It is 5 times as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q2/5

Write an expression for "subtract 4 from 20, then divide by 2."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q3/5

1021

QUESTION ID

S28-B2-Q3

TYPE

Short answer

CORRECT ANSWER

(20 − 4) ÷ 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q3/5

True or false?

(7 + 8) × 3 is three times as large as 7 + 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q4/5

1022

QUESTION ID

S28-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$10 \times (5 - 2)$ is 10 more than $5 - 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q5/5

1023

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • 7.EE.A.2

FORWARD 1

Write equivalent expressions with the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$4^{5} \times 4^{-2}$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q1/5

1025

QUESTION ID

S28-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4^{3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write as a single power.

$(2^3)^4$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q2/5

1026

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2^{12}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q2/5

Which number is equal to $5^6 \div 5^8$?

A. 25

B. 5^{14}

C. $\frac{1}{25}$

D. -25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q3/5

1027

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q3/5

True or false?

$7^0 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q4/5

1028

QUESTION ID

S28-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q4/5

True or false?

$3^2 \times 3^3 = 9^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q5/5

1029

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^2 \times 3^3 = 3^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q5/5

SET 28 • 7.EE.A.2

FORWARD 2

Rewrite percent-growth expressions to reveal the growth factor • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3

A town's population after t years is P(1.02)^t.

Rewrite the expression to show the growth factor for each 10 years, rounded to the nearest hundredth.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q1/5

1031

QUESTION ID
S28-F2-Q1

TYPE
Short answer

CORRECT ANSWER
P(1.22)^(t/10); about 22% growth every 10 years

GRADING NOTE P((1.02)^10)^(t/10) with (1.02)^10 ≈ 1.22 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q1/5

A price p is raised 10%, and then the new price is raised 10% again: p(1.1)(1.1).

Write this as a single factor times p, and give the total percent increase.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q2/5

1032

QUESTION ID

S28-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.21p; a 21% increase

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q2/5

A value after t years is 50(1.03)^t.

What is the yearly percent increase?

A. 50%

B. 3%

C. 0.3%

D. 103%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q3/5

1033

QUESTION ID

S28-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q3/5

True or false?

The expression 1,000(0.97)^t shows a value that decreases by 3% each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q4/5

1034

QUESTION ID

S28-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q4/5

True or false?

A 10% increase followed by a 10% decrease leaves a price unchanged.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q5/5

1035

QUESTION ID
S28-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The final price is 0.99 times the original price.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q5/5

SET 29 • 7.EE.B.3 • EXPRESSIONS & EQUATIONS

Solve multistep problems with positive and negative rational numbers in any form

MAIN	Grade 7 main question	**7.EE.B.3**
BACKWARD 1	Interpret a fraction as division	**5.NF.B.3**
BACKWARD 2	Find a percent of a quantity	**6.RP.A.3.c**
FORWARD 1	Solve real-world problems with two linear equations	**8.EE.C.8.c**
FORWARD 2	Represent constraints with equations or inequalities	**HSA-CED.A.3**

SET 29 • 7.EE.B.3

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.3

Jon earns $16.50 per hour. He gets a raise of $\frac{1}{5}$ of his wage. Then he works 30 hours.

How much does he earn for the 30 hours?

Earnings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q1/5

1038

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

$594

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A board is $8\frac{1}{4}$ feet long. Ana cuts off 2.5 feet. She cuts the rest into 3 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q2/5

1039

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

1 11/12 feet

GRADING NOTE 23/12 feet is also correct. The question asks for an exact length, so a rounded decimal is not full credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q2/5

A 60-gallon tank is $\frac{3}{4}$ full. Then 12.5 gallons are used.

How many gallons are left?

A. 57.5

B. 47.5

C. 32.5

D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q3/5

1040

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 32.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q3/5

True or false?

A $40 sweater is 25% off. Mia pays with a $50 bill. Her change is $20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q4/5

1041

QUESTION ID

S29-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q4/5

True or false?

Leo runs $2\frac{1}{2}$ miles on Monday and 3.4 miles on Tuesday. He runs 6.4 miles in all.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q5/5

1042

QUESTION ID
S29-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE He runs 5.9 miles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q5/5

SET 29 • 7.EE.B.3

BACKWARD 1

Interpret a fraction as division • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $\frac{3}{4}$ as a division expression. Then find its decimal value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q1/5

1044

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3 ÷ 4 = 0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q1/5

5 friends share 2 pizzas equally.

How much pizza does each friend get?

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q2/5

1045

QUESTION ID

S29-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2/5 of a pizza

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q2/5

Which number is equal to 7 ÷ 8?

A. 7.8

B. $\frac{7}{8}$

C. $\frac{8}{7}$

D. 0.78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q3/5

1046

QUESTION ID

S29-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q3/5

True or false?

$\frac{9}{2} = 9 \div 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q4/5

1047

QUESTION ID

S29-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q4/5

True or false?

$3 \div 5 = \frac{5}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q5/5

1048

QUESTION ID

S29-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 7.EE.B.3

BACKWARD 2

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 35% of 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q1/5

1050

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q1/5

What is 12.5% of 64?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q2/5

1051

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q2/5

What is 150% of 20?

A. 130

B. 30

C. 3

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q3/5

1052

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q3/5

True or false?

60% of 25 is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q4/5

1053

QUESTION ID

S29-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

5% of 300 is 1.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q5/5

1054

QUESTION ID

S29-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5% of 300 is 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q5/5

SET 29 • 7.EE.B.3

FORWARD 1

Solve real-world problems with two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.c

Adult tickets cost $9 and child tickets cost $5. Someone buys 20 tickets for $140.

How many adult tickets were bought?

Adult tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q1/5

1056

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q1/5

Plan A costs $20 plus $0.10 per minute. Plan B costs $0.35 per minute.

For how many minutes do the plans cost the same?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q2/5

1057

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q2/5

Two numbers have a sum of 30 and a difference of 8.

What is the larger number?

A. 15

B. 11

C. 19

D. 22

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q3/5

1058

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q3/5

True or false?

$x = 3$ and $y = 5$ is a solution of both $x + y = 8$ and $2x - y = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q4/5

1059

QUESTION ID

S29-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q4/5

True or false?

$x = 4$ and $y = 2$ is a solution of both $x + y = 6$ and $x - y = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q5/5

1060

QUESTION ID

S29-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q5/5

SET 29 • 7.EE.B.3

FORWARD 2

Represent constraints with equations or inequalities • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A baker has 24 cups of flour. Each muffin uses $\frac{3}{4}$ cup and each loaf uses 2.5 cups.

Write an inequality for m muffins and l loaves.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q1/5

1062

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.75m + 2.5l ≤ 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q1/5

A baker has 24 cups of flour. Each muffin uses $\frac{3}{4}$ cup and each loaf uses 2.5 cups.

Can she make 12 muffins and 7 loaves? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q2/5

1063

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No. 12 muffins and 7 loaves need 9 + 17.5 = 26.5 cups, which is more than 24 cups.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q2/5

A trip budget is at most $150. Gas costs $3.50 per gallon (g) and meals cost $12 each (f).

Which inequality shows the budget?

A. $3.5g + 12f \leq 150$

B. $12g + 3.5f \leq 150$

C. $3.5 + 12 \leq 150gf$

D. $3.5g + 12f \geq 150$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S29-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q3/5

1064

QUESTION ID

S29-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3.5g + 12f ≤ 150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S29-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q3/5

True or false?

m = 8 and l = 4 makes 0.75m + 2.5l ≤ 24 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q4/5

1065

QUESTION ID

S29-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q4/5

True or false?

m = 10 and l = 9 makes 0.75m + 2.5l ≤ 24 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q5/5

1066

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.75(10) + 2.5(9) = 30, which is more than 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q5/5

SET 30 • 7.EE.B.3 • EXPRESSIONS & EQUATIONS

Use estimation to judge whether an answer is reasonable

MAIN	Grade 7 main question	**7.EE.B.3**
BACKWARD 1	Round decimals	**5.NBT.A.4**
BACKWARD 2	Use rounding to check the answer to a two-step word problem	**4.OA.A.3**
FORWARD 1	Estimate very large or very small quantities with powers of 10	**8.EE.A.3**
FORWARD 2	Define appropriate quantities and units for a model	**HSN-Q.A.2**

SET 30 • 7.EE.B.3

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.3

Estimate 19.8 × 4.1 by rounding each number to the nearest whole number.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q1/5

1069

QUESTION ID

S30-M-Q1

TYPE

Short answer

CORRECT ANSWER

80

GRADING NOTE 20 × 4 = 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q1/5

Kim says 48.7 + 31.2 = 799.

Use estimation to explain whether her answer is reasonable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q2/5

1070

QUESTION ID

S30-M-Q2

TYPE

Short answer

CORRECT ANSWER

Not reasonable. 49 + 31 is about 80, so the sum should be about 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q2/5

A meal costs $38.75. Ana adds a 20% tip.

Which is the best estimate of the total?

A. About $40

B. About $58

C. About $78

D. About $47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q3/5

1071

QUESTION ID

S30-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. About $47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q3/5

True or false?

$6\frac{7}{8} \times 3\frac{1}{10}$ is about 7 × 3, so a product of about 21 is reasonable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q4/5

1072

QUESTION ID

S30-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q4/5

True or false?

A $59.99 jacket is 30% off. A sale price of about $18 is reasonable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q5/5

1073

QUESTION ID

S30-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sale price should be about $42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q5/5

SET 30 • 7.EE.B.3

BACKWARD 1

Round decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Round 7.86 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q1/5

1075

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 12.348 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q2/5

1076

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q2/5

Round 4.165 to the nearest hundredth.

A. 4.17

B. 4.1

C. 4.2

D. 4.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q3/5

1077

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4.17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q3/5

True or false?

3.97 rounded to the nearest tenth is 4.0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q4/5

1078

QUESTION ID

S30-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q4/5

True or false?

15.52 rounded to the nearest whole number is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q5/5

1079

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 7.EE.B.3

BACKWARD 2

Use rounding to check the answer to a two-step word problem • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A store sells 4 packs of cards for $52 each and charges $11 for shipping.

Round 52 to the nearest ten, and keep 4 and 11 unchanged, to estimate the total cost.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q1/5

1081

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$211

GRADING NOTE $4 \times 50 + 11 = 211$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A store sells 4 packs of cards for $52 each and charges $11 for shipping. Ty says the total is $2,019.

Round 52 to the nearest ten to decide whether his answer is reasonable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q2/5

1082

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Not reasonable. 4 × 50 + 11 = 211, so the total should be about $211.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q2/5

A school buys 6 boxes of 48 pencils and gives away 95 pencils.

Round 48 and 95 to the nearest ten, and keep 6 unchanged. What is the best estimate of the pencils left?

A. 300

B. 200

C. 400

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q3/5

1083

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q3/5

True or false?

A farm picks 312 apples on Monday and 289 apples on Tuesday, then sells 198 apples. Rounding each number to the nearest hundred gives an estimate of 400 apples left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q4/5

1084

QUESTION ID

S30-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q4/5

True or false?

A bus carries 38 riders on each of 7 trips and 21 riders on one more trip. Rounding 38 and 21 to the nearest ten, and keeping 7 unchanged, gives an estimate of 220 riders.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q5/5

1085

QUESTION ID
S30-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The estimate is 7 × 40 + 20 = 300 riders.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 7.EE.B.3

FORWARD 1

Estimate very large or very small quantities with powers of 10 • Grade 8 • Expressions & Equations • 8.EE.A.3

Estimate 38,900,000 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q1/5

1087

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4×10^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q1/5

Estimate 0.00062 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q2/5

1088

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6×10^{-4}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q2/5

About how many times as large as 3×10^5 is 9×10^8?

A. 300

B. 3

C. 30

D. 3,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q3/5

1089

QUESTION ID

S30-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q3/5

True or false?

3×10^8 is a reasonable estimate for 296,000,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q4/5

1090

QUESTION ID

S30-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

6×10^{-3} is about 6,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q5/5

1091

QUESTION ID

S30-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $6 \times 10^{-3} = 0.006$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q5/5

SET 30 • 7.EE.B.3

FORWARD 2

Define appropriate quantities and units for a model • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2

You want to estimate how many times a heart beats in one day.

Which two quantities do you need, and what are their units?

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q1/5

1093

QUESTION ID

S30-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The heart rate (beats per minute) and the number of minutes in a day (minutes per day).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q1/5

You want to model the cost of gas for a car trip to another city.

Name three quantities you would define, with their units.

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q2/5

1094

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The distance (miles), the car's fuel efficiency (miles per gallon), and the gas price (dollars per gallon).

GRADING NOTE Any three relevant quantities with sensible units are correct; for example, kilometers and liters may be used instead.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q2/5

Which quantity is most useful for estimating the cost to paint the walls of a room?

A. The volume of the room

B. The height of the door

C. The number of windows

D. The area of the walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q3/5

1095

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The area of the walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q3/5

True or false?

Gallons of water used per day is a reasonable quantity for modeling a family's water use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q4/5

1096

QUESTION ID

S30-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q4/5

True or false?

The color of a car is a useful quantity for modeling how much fuel it uses on a trip.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q5/5

1097

QUESTION ID

S30-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q5/5

SET 31 • 7.EE.B.4.a • EXPRESSIONS & EQUATIONS

Solve word problems that lead to equations of the form px + q = r

MAIN	Grade 7 main question	**7.EE.B.4.a**
BACKWARD 1	Solve one-step equations	**6.EE.B.7**
BACKWARD 2	Write an expression with a variable for a situation	**6.EE.B.6**
FORWARD 1	Solve equations with the variable on both sides	**8.EE.C.7.b**
FORWARD 2	Rearrange a formula to solve for a chosen quantity	**HSA-CED.A.4**

SET 31 • 7.EE.B.4.a

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges a $25 sign-up fee plus $18 per month. Ty has paid $151 in all.

Write and solve an equation to find the number of months m.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q1/5

1100

QUESTION ID

S31-M-Q1

TYPE

Short answer

CORRECT ANSWER

The equation 18m + 25 = 151 and the answer m = 7 months

GRADING NOTE Both parts are required: a valid equation (for example 18m + 25 = 151) and the correct answer, 7 months. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q1/5

Solve for x.

$4x + 9 = -11$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q2/5

1101

QUESTION ID

S31-M-Q2

TYPE

Short answer

CORRECT ANSWER

−5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q2/5

Ava has $60. She buys movie tickets for $7.50 each and has $15 left.

Which equation finds the number of tickets t?

A. $60t - 7.5 = 15$

B. $7.5 - 60t = 15$

C. $7.5t + 60 = 15$

D. $60 - 7.5t = 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q3/5

1102

QUESTION ID

S31-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60 − 7.5t = 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q3/5

True or false?

$x = 6$ is the solution of $\frac{1}{2}x + 3 = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q4/5

1103

QUESTION ID

S31-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q4/5

True or false?

A plumber charges $45 plus $30 per hour. A bill of $165 means the plumber worked 5 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q5/5

1104

QUESTION ID

S31-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The plumber worked 4 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q5/5

SET 31 • 7.EE.B.4.a

BACKWARD 1

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

x + 13 = 41

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q1/5

1106

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for y.

9y = 72

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q2/5

1107

QUESTION ID

S31-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for n.

$n - 6.4 = 3.6$

A. n = 10

B. n = 2.8

C. n = −2.8

D. n = 23.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q3/5

1108

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. n = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q3/5

True or false?

k = 7 is the solution of 5k = 35.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q4/5

1109

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q4/5

True or false?

p = 9 is the solution of p + 4 = 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q5/5

1110

QUESTION ID

S31-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • 7.EE.B.4.a

BACKWARD 2

Write an expression with a variable for a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

A ticket costs $12.

Write an expression for the cost of t tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q1/5

1112

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q1/5

Ana has 5 more stickers than Ben. Ben has b stickers.

Write an expression for the number of stickers Ana has.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q2/5

1113

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

b + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q2/5

A pizza has s slices. 4 friends share it equally.

Which expression gives the number of slices each friend gets?

A. $s - 4$

B. $4s$

C. $s + 4$

D. $\frac{s}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q3/5

1114

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. s/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q3/5

True or false?

A pen costs p dollars. 3 pens cost 3p dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q4/5

1115

QUESTION ID

S31-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q4/5

True or false?

Mo is m years old. His age in 6 years is 6m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q5/5

1116

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE His age in 6 years is m + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • 7.EE.B.4.a

FORWARD 1

Solve equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A charges $40 plus $5 per day. Store B charges $25 plus $8 per day.

After how many days do the stores charge the same amount?

Days: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q1/5

1118

QUESTION ID

S31-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q1/5

Solve for x.

$7x - 4 = 3x + 20$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q2/5

1119

QUESTION ID

S31-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$-2x + 9 = 4x - 15$

A. $x = -4$

B. $x = 1$

C. $x = 4$

D. $x = 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q3/5

1120

QUESTION ID

S31-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q3/5

True or false?

$x = -1$ is the solution of $5x + 3 = 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q4/5

1121

QUESTION ID

S31-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = 2$ is the solution of $6x - 1 = 4x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q5/5

1122

QUESTION ID

S31-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q5/5

SET 31 • 7.EE.B.4.a

FORWARD 2

Rearrange a formula to solve for a chosen quantity • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve P = 2l + 2w for l.

l = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q1/5

1124

QUESTION ID

S31-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(P − 2w)/2

GRADING NOTE l = P/2 − w is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve y = mx + b for x. Assume m ≠ 0.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q2/5

1125

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(y − b)/m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q2/5

Solve $A = \frac{1}{2}bh$ for h, where the base $b > 0$.

A. $h = 2Ab$

B. $h = \frac{A}{2b}$

C. $h = \frac{b}{2A}$

D. $h = \frac{2A}{b}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q3/5

1126

QUESTION ID

S31-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. h = 2A/b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q3/5

True or false?

For a speed $r > 0$, solving $d = rt$ for t gives $t = \frac{d}{r}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q4/5

1127

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q4/5

True or false?

Solving C = 5n + 20 for n gives n = 5C − 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q5/5

1128

QUESTION ID
S31-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE n = (C − 20)/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q5/5

SET 32 • 7.EE.B.4.a • EXPRESSIONS & EQUATIONS

Solve word problems that lead to equations of the form p(x + q) = r

MAIN	Grade 7 main question	**7.EE.B.4.a**
BACKWARD 1	Evaluate expressions with parentheses	**5.OA.A.1**
BACKWARD 2	Use the distributive property to expand an expression	**6.EE.A.3**
FORWARD 1	Solve equations with the distributive property on both sides	**8.EE.C.7.b**
FORWARD 2	Solve quadratic equations by square roots or factoring	**HSA-REI.B.4.b**

SET 32 • 7.EE.B.4.a

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle is 7 cm wide and (x + 3) cm long. Its area is 63 cm².

Write and solve an equation to find x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q1/5

1131

QUESTION ID

S32-M-Q1

TYPE

Short answer

CORRECT ANSWER

The equation 7(x + 3) = 63 and the answer x = 6

GRADING NOTE Both parts are required: a valid equation (for example 7(x + 3) = 63) and the correct answer, x = 6. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q1/5

Solve for x.

$-3(x - 4) = 27$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q2/5

1132

QUESTION ID

S32-M-Q2

TYPE

Short answer

CORRECT ANSWER

−5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q2/5

5 friends each buy a $4 snack and a drink that costs d dollars. They spend $32.50 in all.

Which equation finds d?

A. d + 5(4) = 32.5

B. 5(d + 4) = 32.5

C. 5d + 4 = 32.5

D. 4(d + 5) = 32.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q3/5

1133

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5(d + 4) = 32.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q3/5

True or false?

x = 2.5 is the solution of 2(x + 1.5) = 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q4/5

1134

QUESTION ID

S32-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q4/5

True or false?

The solution of $\frac{1}{3}(x - 6) = 4$ is $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q5/5

1135

QUESTION ID

S32-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q5/5

SET 32 • 7.EE.B.4.a

BACKWARD 1

Evaluate expressions with parentheses • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate.

$3 \times (8 + 4)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q1/5

1137

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate.

$(15 - 6) \times 2$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q2/5

1138

QUESTION ID

S32-B1-Q2

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate.

$4 \times (9 - 2) + 1$

A. 25

B. 29

C. 36

D. 33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q3/5

1139

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q3/5

True or false?

$2 \times (5 + 5) = 20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q4/5

1140

QUESTION ID

S32-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q4/5

True or false?

$(12 + 3) \times 2 = 18$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q5/5

1141

QUESTION ID

S32-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 7.EE.B.4.a

BACKWARD 2

Use the distributive property to expand an expression • Grade 6 • Expressions & Equations • 6.EE.A.3

Rewrite using the distributive property.

9(m + 2)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q1/5

1143

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9m + 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite using the distributive property.

4(3k + 5)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q2/5

1144

QUESTION ID

S32-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12k + 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q2/5

Which expression is equivalent to 6(x + 1.5)?

A. 6x + 1.5

B. 6x + 9

C. 6x + 7.5

D. 7x + 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q3/5

1145

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q3/5

True or false?

$5(2 + y) = 10 + 5y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q4/5

1146

QUESTION ID

S32-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q4/5

True or false?

$8(p + 3) = 8p + 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q5/5

1147

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 7.EE.B.4.a

FORWARD 1

Solve equations with the distributive property on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$3(x + 2) = 2(x + 7)$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q1/5

1149

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q1/5

Solve for x.

$-2(x - 5) = 4(x + 1)$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q2/5

1150

QUESTION ID

S32-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q2/5

Solve for x.

$\frac{1}{2}(6x - 4) = 2(x + 3)$

A. x = 2

B. x = 8

C. x = −8

D. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q3/5

1151

QUESTION ID

S32-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q3/5

True or false?

$x = 4$ is the solution of $5(x - 1) = 3(x + 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q4/5

1152

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q4/5

True or false?

$x = 3$ is the solution of $2(x + 6) = 4(x - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q5/5

1153

QUESTION ID

S32-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q5/5

SET 32 • 7.EE.B.4.a

FORWARD 2

Solve quadratic equations by square roots or factoring • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x. Give all solutions.

$(x + 3)^2 = 49$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q1/5

1155

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

x = 4 and x = −10

GRADING NOTE Both solutions are required. Either order and set notation, such as {4, −10}, are accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x. Give all solutions.

$x^2 = 81$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q2/5

1156

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 9 and x = −9

GRADING NOTE Both solutions are required; x = ±9 and set notation are accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q2/5

Solve for x.

$(x - 2)(x + 5) = 0$

A. $x = -5$ only

B. $x = 2$ only

C. $x = -2$ or $x = 5$

D. $x = 2$ or $x = -5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q3/5

1157

QUESTION ID

S32-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2 or x = −5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q3/5

True or false?

The solutions of $2(x - 1)^2 = 18$ are $x = 4$ and $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q4/5

1158

QUESTION ID

S32-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q4/5

True or false?

$x^2 = 16$ has only one solution, $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q5/5

1159

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 4$ or $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q5/5

SET 33 • 7.EE.B.4.a • EXPRESSIONS & EQUATIONS

Compare an algebraic solution with an arithmetic solution

MAIN	Grade 7 main question	**7.EE.B.4.a**
BACKWARD 1	Solve two-step word problems	**3.OA.D.8**
BACKWARD 2	Decide whether a value makes an equation true	**6.EE.B.5**
FORWARD 1	Solve systems of two linear equations algebraically and by inspec…	**8.EE.C.8.b**
FORWARD 2	Justify a complete solution and find an invalid step	**HSA-REI.A.1**

SET 33 • 7.EE.B.4.a

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges $15 to join plus $8 for each class. Max paid $79 in all.

(a) Solve with arithmetic. (b) Write and solve an equation. (c) Compare the steps of the two solutions.

Classes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q1/5

1162

QUESTION ID

S33-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

8 classes

GRADING NOTE REQUIRED WORK: Grade all three parts: (a) a correct arithmetic solution, (b) a correct equation with an algebraic solution, and (c) a comparison that matches each arithmetic operation to the algebra step that does the same thing in the same order. Example: (a) 79 − 15 = 64, 64 ÷ 8 = 8; (b) 8c + 15 = 79, subtract 15 from both sides, then divide both sides by 8; (c) both subtract 15 first and then divide by 8. Any valid equation is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Four friends each buy a ticket and a $3 popcorn. They pay $46 in all.

(a) Solve with arithmetic. (b) Write and solve an equation. (c) Compare the steps of the two solutions.

Ticket price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q2/5

1163

QUESTION ID

S33-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$8.50

GRADING NOTE REQUIRED WORK: Grade all three parts: (a) a correct arithmetic solution, (b) a correct equation with an algebraic solution, and (c) a comparison that matches each arithmetic operation to the algebra step that does the same thing in the same order. Example: (a) 46 ÷ 4 = 11.50, 11.50 − 3 = 8.50; (b) 4(t + 3) = 46, divide both sides by 4, then subtract 3; (c) both divide by 4 first and then subtract 3. An equation such as 4t + 12 = 46 is also accepted, with the matching comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q2/5

An algebraic solution of $5x - 8 = 27$ adds 8 to both sides and then divides both sides by 5.

Which arithmetic steps match it?

A. Multiply 27 by 5, then add 8.

B. Add 8 to 27, then divide by 5.

C. Divide 27 by 5, then add 8.

D. Subtract 8 from 27, then divide by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q3/5

1164

QUESTION ID

S33-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Add 8 to 27, then divide by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An arithmetic solution of $3(x + 4) = 33$ is $33 \div 3 - 4$.

For each arithmetic operation, name the algebra step that matches it. Then give x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q4/5

1165

QUESTION ID

S33-M-Q4

TYPE

Short answer

CORRECT ANSWER

Dividing 33 by 3 matches dividing both sides by 3; subtracting 4 matches subtracting 4 from both sides; x = 7.

GRADING NOTE Grade both matches and the value x = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q4/5

True or false?

An equation multiplies x by 2 and then adds 6: 2x + 6 = 20. The arithmetic (20 − 6) ÷ 2 undoes those operations in reverse order, just as the algebra steps do.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q5/5

1166

QUESTION ID

S33-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q5/5

SET 33 • 7.EE.B.4.a

BACKWARD 1

Solve two-step word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

Jo has 45 stickers. She gives away 9. Then she shares the rest equally among 4 friends.

How many stickers does each friend get?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q1/5

1168

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q1/5

There are 6 boxes with 8 pens in each box. 5 pens are lost.

How many pens are left?

Pens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q2/5

1169

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

43

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q2/5

A book has 90 pages. Kim reads 20 pages on Monday and 35 pages on Tuesday.

How many pages are left?

A. 15

B. 55

C. 145

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q3/5

1170

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q3/5

True or false?

3 packs of 7 cards and 4 more cards make 25 cards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q4/5

1171

QUESTION ID

S33-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q4/5

True or false?

40 apples are shared equally in 5 bags. Then 2 apples are taken out of each bag. Each bag now has 8 apples.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q5/5

1172

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each bag has 6 apples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • 7.EE.B.4.a

BACKWARD 2

Decide whether a value makes an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 4 a solution of 3x + 2 = 14? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q1/5

1174

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 3(4) + 2 = 14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q1/5

Which value of n makes n + 9 = 15 true?

A. 9

B. 24

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q2/5

1175

QUESTION ID

S33-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q2/5

Which of the values 2, 5, and 8 makes 7y = 35 true?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q3/5

1176

QUESTION ID

S33-B2-Q3

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q3/5

True or false?

m = 3 makes 12 ÷ m = 4 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q4/5

1177

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q4/5

True or false?

a = 5 makes 2a − 1 = 11 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q5/5

1178

QUESTION ID

S33-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 7.EE.B.4.a

FORWARD 1

Solve systems of two linear equations algebraically and by inspection • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Solve the system.

$y = 2x$

$x + y = 18$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q1/5

1180

QUESTION ID

S33-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 6, y = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q1/5

Solve the system.

$y = x + 3$

$y = -2x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q2/5

1181

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 2, y = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q2/5

Look at the system. How many solutions does it have?

$3x + 2y = 5$

$3x + 2y = 6$

A. Infinitely many solutions

B. Two solutions

C. No solution

D. One solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q3/5

1182

QUESTION ID

S33-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q3/5

True or false?

The system y = 4x and y = 4x + 1 has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q4/5

1183

QUESTION ID

S33-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = 4$ and $y = 6$ is the solution of the system $x + y = 10$ and $x - y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q5/5

1184

QUESTION ID

S33-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 6$, $y = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q5/5

SET 33 • 7.EE.B.4.a

FORWARD 2

Justify a complete solution and find an invalid step • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2(x + 3) = \frac{1}{2}(4x - 6) + x$. Write each step and the reason for it.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q1/5

1186

QUESTION ID

S33-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

9

GRADING NOTE REQUIRED WORK: Every step with a valid reason. Example: distributive property on both sides ($2x + 6 = 2x - 3 + x$); combine like terms ($2x + 6 = 3x - 3$); subtract $2x$ from both sides ($6 = x - 3$); add 3 to both sides ($x = 9$). Other valid orders are accepted. A correct answer without reasons is not full credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q1/5

Mo solved $\frac{x}{2} + 3 = x - 1$ like this:

Step 1: $x + 3 = 2x - 1$

Step 2: $x = 4$

Which step is invalid? Correct it and give the solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q2/5

1187

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Step 1: multiplying both sides by 2 gives $x + 6 = 2x - 2$, so $x = 8$.

GRADING NOTE Grade both: Step 1 with the correction, and $x = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q2/5

Which reason justifies rewriting $4x - 7 = 2x + 5$ as $2x - 7 = 5$?

A. Commutative property of multiplication

B. Distributive property

C. Subtraction property of equality (subtract 2x from both sides)

D. Division property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q3/5

1188

QUESTION ID

S33-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Subtraction property of equality (subtract 2x from both sides)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q3/5

True or false?

Dividing both sides of $3(x - 2) = 3x - 6$ by 3 gives $x - 2 = x - 2$, so every number is a solution of the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q4/5

1189

QUESTION ID

S33-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q4/5

True or false?

Dividing both sides of $x^2 = 5x$ by x keeps every solution of the equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q5/5

1190

QUESTION ID
S33-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It loses the solution $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q5/5

SET 34 • 7.EE.B.4.b • EXPRESSIONS & EQUATIONS

Solve word problems that lead to inequalities of the form px + q > r or px + q < r

MAIN	Grade 7 main question	**7.EE.B.4.b**
BACKWARD 1	Write an inequality for a situation	**6.EE.B.8**
BACKWARD 2	Decide whether a value makes an inequality true	**6.EE.B.5**
FORWARD 1	Solve multistep equations with the variable on both sides — near…	**8.EE.C.7.b**
FORWARD 2	Solve linear inequalities with the variable on both sides	**HSA-REI.B.3**

SET 34 • 7.EE.B.4.b

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Sam has $40. He buys a $12 shirt and some pairs of socks that cost $3.50 each.

Write and solve an inequality for the number of pairs of socks p he can buy.

Inequality and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q1/5

1193

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

The inequality $3.5p + 12 \leq 40$ and the solution $p \leq 8$: he can buy 0 to 8 pairs (whole numbers).

GRADING NOTE Both parts are required: a valid inequality and its solution, stated as whole numbers of pairs from 0 to 8. Any valid inequality is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plumber charges $55 plus $40 per hour. Ana can spend no more than $255.

Write and solve an inequality for the number of hours h.

Inequality and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q2/5

1194

QUESTION ID

S34-M-Q2

TYPE

Short answer

CORRECT ANSWER

The inequality 40h + 55 ≤ 255 and the solution h ≤ 5: at most 5 hours.

GRADING NOTE Both parts are required. Time can be any amount from 0 to 5 hours. Any valid inequality is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q2/5

A club has $120 and earns $15 for each car it washes. It needs more than $300.

Which inequality and solution fit the problem?

A. 120c + 15 > 300; c > 2.375

B. 15c > 300 + 120; c > 28

C. 15c + 120 > 300; c > 12

D. 15c + 120 < 300; c < 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q3/5

1195

QUESTION ID

S34-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 15c + 120 > 300; c > 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q3/5

True or false?

A tank holds 500 liters and drains 25 liters per minute. The water stays above 200 liters while $500 - 25t > 200$, which is while $t < 12$ minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q4/5

1196

QUESTION ID

S34-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q4/5

True or false?

Kim has $20 and saves $6 each week. To have at least $80, she solves 6w + 20 ≥ 80, so she needs at most 10 weeks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q5/5

1197

QUESTION ID
S34-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE w ≥ 10: she needs at least 10 weeks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q5/5

SET 34 • 7.EE.B.4.b

BACKWARD 1

Write an inequality for a situation • Grade 6 • Expressions & Equations • 6.EE.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Riders must be at least 48 inches tall.

Write an inequality for a rider's height h.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q1/5

1199

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

h ≥ 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q1/5

Write an inequality for "a temperature t below −2°F."

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q2/5

1200

QUESTION ID

S34-B1-Q2

TYPE

Short answer

CORRECT ANSWER

t < −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q2/5

Which inequality means "x is more than 7"?

A. $x < 7$

B. $x > 7$

C. $x \leq 7$

D. $x = 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q3/5

1201

QUESTION ID

S34-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x > 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q3/5

True or false?

$x < 3$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q4/5

1202

QUESTION ID

S34-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q4/5

True or false?

5 is a solution of $x < 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q5/5

1203

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • 7.EE.B.4.b

BACKWARD 2

Decide whether a value makes an inequality true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is 9 a solution of 2x > 16? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q1/5

1205

QUESTION ID

S34-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 2(9) = 18, and 18 > 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q1/5

Which value makes y + 4 < 10 true?

A. 8

B. 10

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q2/5

1206

QUESTION ID

S34-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which of the values 3, 6, and 9 make $n - 2 \geq 4$ true?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q3/5

1207

QUESTION ID

S34-B2-Q3

TYPE

Short answer

CORRECT ANSWER

6 and 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q3/5

True or false?

x = 0 makes 3x < 1 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q4/5

1208

QUESTION ID

S34-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q4/5

True or false?

k = 2 makes 5k > 10 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q5/5

1209

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • 7.EE.B.4.b

FORWARD 1

Solve multistep equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$0.5x + 7 = 2x - 2$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q1/5 • NEAREST RELATED

1211

QUESTION ID

S34-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$\frac{2}{3}(x + 3) = x - 1$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q2/5 • NEAREST RELATED

1212

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q2/5 • NEAREST RELATED

Solve for x.

$6 - 3x = x + 22$

A. $x = -4$

B. $x = 7$

C. $x = -7$

D. $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q3/5 • NEAREST RELATED

1213

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

$x = -2$ is the solution of $4(x + 5) = x + 14$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q4/5 • NEAREST RELATED

1214

QUESTION ID

S34-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

$x = 1$ is the solution of $9x - 3 = 2x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q5/5 • NEAREST RELATED

1215

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 5/7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 34 • 7.EE.B.4.b

FORWARD 2

Solve linear inequalities with the variable on both sides • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve.

$5x - 3 > 2x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q1/5

1217

QUESTION ID

S34-F2-Q1

TYPE

Short answer

CORRECT ANSWER

x > 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q1/5

Solve.

$-4(x + 1) \leq 2x + 8$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q2/5

1218

QUESTION ID

S34-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x \geq -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q2/5

Solve.

$7 - 2x < x - 8$

A. $x < -5$

B. $x > -5$

C. $x > 5$

D. $x < 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q3/5

1219

QUESTION ID

S34-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x > 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q3/5

True or false?

$x = 0$ is a solution of $3(x - 2) < x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q4/5

1220

QUESTION ID

S34-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q4/5

True or false?

The solution of $6x + 1 \geq 4x + 11$ is $x \leq 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q5/5

1221

QUESTION ID

S34-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x \geq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q5/5

SET 35 • 7.EE.B.4.b • EXPRESSIONS & EQUATIONS

Graph the solution set of an inequality and interpret it in context

MAIN	Grade 7 main question	**7.EE.B.4.b**
BACKWARD 1	Graph an inequality on a number line	**6.EE.B.8**
BACKWARD 2	Read an inequality as positions on a number line	**6.NS.C.7.a**
FORWARD 1	Find the solution of a system of equations from its graph	**8.EE.C.8.a**
FORWARD 2	Graph the solutions of a linear inequality in two variables	**HSA-REI.D.12**

SET 35 • 7.EE.B.4.b

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

A taxi charges $3 plus $2 per mile. Ty can spend at most $15.

Solve 2m + 3 ≤ 15 for the miles m. Then graph the possible distances on the number line. (Distances cannot be negative.)



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q1/5

1224

QUESTION ID

S35-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A closed circle at 6 and the line shaded from 0 to 6 (a closed circle or the end of the shading at 0)

Written: m ≤ 6, so 0 ≤ m ≤ 6 miles

GRADING NOTE Grade both: the solution 0 ≤ m ≤ 6 (m ≤ 6 with distances not negative), and the graph from 0 to 6 with a closed circle at 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A freezer is at 2°C and cools 1.5°C each hour.

Solve $2 - 1.5t < -4$ to find when it is colder than −4°C. Then graph the solution on the number line.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q2/5

1225

QUESTION ID

S35-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: An open circle at 4 with an arrow or shading to the right

Written: t > 4: after more than 4 hours

GRADING NOTE Grade both: t > 4 and the graph with an open circle at 4 shaded to the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q2/5

Tickets cost $6 each. Jo has $40 and must keep at least $10, so $6t \le 30$. The graph shows the solution as whole numbers.

Which description fits the graph?



A. Jo can buy any number of tickets greater than 5.

B. Jo can buy 2.5 tickets.

C. Jo can buy exactly 5 tickets.

D. Jo can buy 0, 1, 2, 3, 4, or 5 tickets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q3/5

1226

QUESTION ID

S35-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Jo can buy 0, 1, 2, 3, 4, or 5 tickets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A club has $30 and earns $5 for each ticket it sells. It needs more than $230.

Solve 5t + 30 > 230. Then graph the possible numbers of tickets (whole numbers) on the number line.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q4/5

1227

QUESTION ID

S35-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Dots at the whole numbers 41, 42, 43, ... to the end of the line, with an arrow or "..." to show the pattern continues; no dot at 40

Written: t > 40: the club must sell at least 41 tickets.

GRADING NOTE Grade both: the solution with the context (at least 41 tickets), and the graph of whole numbers greater than 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q4/5

True or false?

A plant is 5 cm tall and grows 2 cm per week. The graph shows the weeks w when the plant is taller than 11 cm, the solution of 2w + 5 > 11.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q5/5

1228

QUESTION ID

S35-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

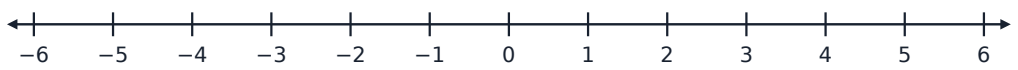
Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q5/5

SET 35 • 7.EE.B.4.b

BACKWARD 1

Graph an inequality on a number line • Grade 6 • Expressions & Equations • 6.EE.B.8

Graph x < 4 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q1/5

1230

QUESTION ID

S35-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

An open circle at 4 with an arrow or shading to the left

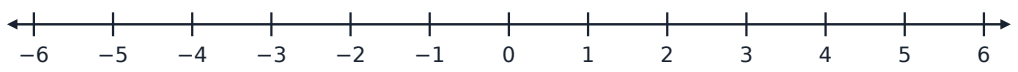
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph x ≥ −1 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q2/5

1231

QUESTION ID

S35-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A closed circle at −1 with an arrow or shading to the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which inequality does the graph show?



A. $x < 0$

B. $x \geq 0$

C. $x \leq 0$

D. $x > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q3/5

1232

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x < 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q3/5

True or false?

A closed circle on the graph of an inequality shows that the endpoint is a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q4/5

1233

QUESTION ID

S35-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of $x > 5$ has a closed circle at 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q5/5

1234

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • 7.EE.B.4.b

BACKWARD 2

Read an inequality as positions on a number line • Grade 6 • The Number System • 6.NS.C.7.a

−4 is to the left of 1 on a number line.

Write an inequality that shows this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q1/5

1236

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−4 < 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q1/5

Which statement says that −2 is to the right of −6 on a number line?

A. −2 < −6

B. −2 = −6

C. −6 > −2

D. −2 > −6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q2/5

1237

QUESTION ID

S35-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. −2 > −6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q2/5

What does $-8 < -3$ tell you about the positions of -8 and -3 on a number line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q3/5

1238

QUESTION ID

S35-B2-Q3

TYPE

Short answer

CORRECT ANSWER

-8 is to the left of -3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q3/5

True or false?

$-1 > -9$ because -1 is to the right of -9 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q4/5

1239

QUESTION ID

S35-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q4/5

True or false?

$-5 > 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q5/5

1240

QUESTION ID

S35-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 7.EE.B.4.b

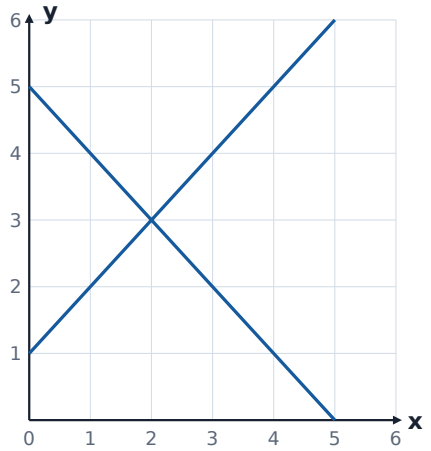
FORWARD 1

Find the solution of a system of equations from its graph • Grade 8 • Expressions & Equations • 8.EE.C.8.a

The graph shows y = x + 1 and y = −x + 5.

At what point do the lines intersect?

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q1/5

1242

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q1/5

Two lines are graphed on the same grid.

What does their point of intersection represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q2/5

1243

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The pair (x, y) that makes both equations true; the solution of the system.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q2/5

The graphs of two linear equations are parallel lines.

How many solutions does the system have?

A. One solution

B. Two solutions

C. Infinitely many solutions

D. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q3/5

1244

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q3/5

True or false?

(1, 4) is the solution of the system y = 2x + 2 and y = 3x + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q4/5

1245

QUESTION ID

S35-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q4/5

True or false?

If two equations have the same graph, the system has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q5/5

1246

QUESTION ID

S35-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

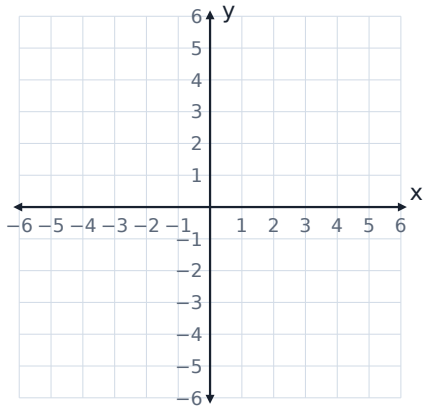
Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q5/5

SET 35 • 7.EE.B.4.b

FORWARD 2

Graph the solutions of a linear inequality in two variables • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

Graph y > x + 1 on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q1/5

1248

QUESTION ID

S35-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A dashed line through (0, 1) with slope 1, with the region above the line shaded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q1/5

Is (2, 5) a solution of y ≥ 2x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q2/5

1249

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. 5 ≥ 2(2), because 5 ≥ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q2/5

Which point is in the solution set of $y < -x + 3$?

A. (0, 0)

B. (0, 5)

C. (3, 1)

D. (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q3/5

1250

QUESTION ID

S35-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q3/5

True or false?

The boundary line of the graph of $y \leq 3x - 2$ is drawn solid.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q4/5

1251

QUESTION ID

S35-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q4/5

True or false?

The solution set of $y > 2$ is the region below the line $y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q5/5

1252

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q5/5

SET 36 • 7.G.A.1 • GEOMETRY

Find actual lengths and areas from a scale drawing

MAIN	Grade 7 main question	**7.G.A.1**
BACKWARD 1	Convert measurement units	**6.RP.A.3.d**
BACKWARD 2	Find the area of a rectangle	**4.MD.A.3**
FORWARD 1	Describe similar figures with dilations	**8.G.A.4**
FORWARD 2	Decide whether two triangles are similar	**HSG-SRT.A.2**

SET 36 • 7.G.A.1

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.1

A scale drawing of a garden uses the scale 1 cm : 2 m. The garden is 6 cm long on the drawing.

What is the actual length of the garden?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q1/5

1255

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

12 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q1/5

A floor plan uses the scale 1 inch : 4 feet. A room is 3 inches by 2.5 inches on the plan.

What is the actual area of the room?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q2/5

1256

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

120 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q2/5

A map uses the scale 1 cm : 25 km. Two towns are 4.4 cm apart on the map.

What is the actual distance between the towns?

A. 100 km

B. 110 km

C. 5.68 km

D. 29.4 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q3/5

1257

QUESTION ID

S36-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 110 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q3/5

True or false?

A drawing uses the scale 1 cm : 3 m. A 5 cm by 2 cm rectangle on the drawing has an actual area of 90 m².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q4/5

1258

QUESTION ID

S36-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q4/5

True or false?

A drawing uses the scale 1 cm : 10 cm. The actual area is 10 times the area on the drawing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q5/5

1259

QUESTION ID

S36-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The actual area is 100 times the area on the drawing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q5/5

SET 36 • 7.G.A.1

BACKWARD 1

Convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

ANSWER KEY Reduced copy of the question page, with its answer

How many inches are in 4.5 feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q1/5

1261

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many meters are in 350 centimeters?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q2/5

1262

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q2/5

How many yards are in 27 feet?

A. 24

B. 9

C. 3

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q3/5

1263

QUESTION ID

S36-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q3/5

True or false?

2.5 kilometers = 2,500 meters

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q4/5

1264

QUESTION ID

S36-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q4/5

True or false?

6 feet = 60 inches

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q5/5

1265

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6 feet = 72 inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q5/5

SET 36 • 7.G.A.1

BACKWARD 2

Find the area of a rectangle • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 12 m long and 9 m wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q1/5

1267

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

108 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q1/5

A rectangle has an area of 84 square feet. It is 12 feet long.

How wide is it?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q2/5

1268

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q2/5

A rectangle is 15 cm long and 6 cm wide.

What is its area?

A. 96 square centimeters

B. 90 square centimeters

C. 21 square centimeters

D. 42 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q3/5

1269

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 90 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q3/5

True or false?

A square with 7-inch sides has an area of 49 square inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q4/5

1270

QUESTION ID

S36-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q4/5

True or false?

A rectangle 10 m long and 4 m wide has an area of 28 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q5/5

1271

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is 40 square meters; 28 m is its perimeter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • 7.G.A.1

FORWARD 1

Describe similar figures with dilations • Grade 8 • Geometry • 8.G.A.4

A triangle has sides 3, 4, and 5 units. It is dilated by a scale factor of 2.5.

What are the new side lengths?

Side lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q1/5

1273

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7.5, 10, and 12.5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q1/5

A 4-by-6 rectangle is similar to a rectangle that is 10 by w. The 4 side matches the 10 side.

Find w.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q2/5

1274

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q2/5

A figure is dilated by a scale factor of $\frac{1}{3}$. One side is 18 cm long.

How long is the matching side of the image?

A. $18\frac{1}{3}$ cm

B. 3 cm

C. 6 cm

D. 54 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q3/5

1275

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q3/5

True or false?

Dilations keep angle measures the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q4/5

1276

QUESTION ID

S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q4/5

True or false?

A 2-by-3 rectangle and a 4-by-5 rectangle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q5/5

1277

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 7.G.A.1

FORWARD 2

Decide whether two triangles are similar • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has angles of 40°, 60°, and 80°. Triangle DEF has angles of 40°, 60°, and 80°.

Are the triangles similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q1/5

1279

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Their corresponding angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

One triangle has sides 6, 8, and 10. Another has sides 9, 12, and 15.

What is the scale factor from the first triangle to the second?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q2/5

1280

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE 3/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q2/5

Which two triangles must be similar?

A. Two right triangles

B. Two triangles that each have a 5 cm side

C. Two isosceles triangles

D. Two triangles that each have a 50° angle and a 70° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q3/5

1281

QUESTION ID

S36-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Two triangles that each have a 50° angle and a 70° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q3/5

True or false?

Two triangles are similar with a scale factor of 3. Each pair of corresponding sides has a ratio of 3 : 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q4/5

1282

QUESTION ID

S36-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q4/5

True or false?

All rectangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q5/5

1283

QUESTION ID

S36-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q5/5

SET 37 • 7.G.A.1 • GEOMETRY

Reproduce a scale drawing at a different scale

MAIN	Grade 7 main question	**7.G.A.1**
BACKWARD 1	Draw a polygon on the coordinate plane	**6.G.A.3**
BACKWARD 2	Compare the size of a product to the size of one factor	**5.NF.B.5.a**
FORWARD 1	Describe the effect of a dilation on coordinates	**8.G.A.3**
FORWARD 2	Verify the properties of dilations	**HSG-SRT.A.1**

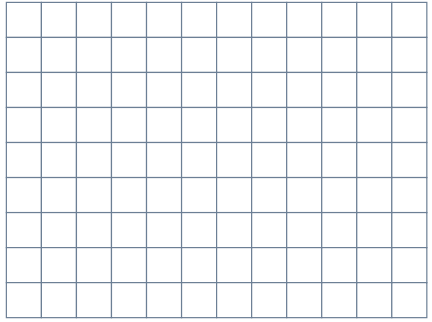
SET 37 • 7.G.A.1

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.1

A scale drawing of a garden is a rectangle 6 units long and 4 units wide. Its scale is 1 unit : 2 m.

Redraw the garden on the grid at a scale of 1 unit : 1 m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q1/5

1286

QUESTION ID

S37-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A rectangle 12 units by 8 units, in any position or orientation

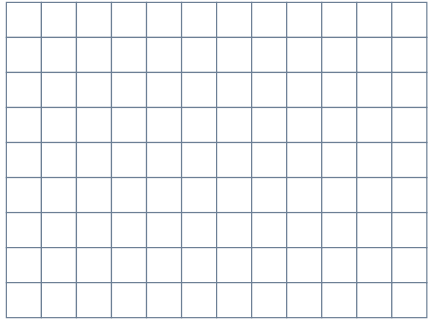
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q1/5

A rectangle on a drawing is 4 units by 2 units.

Draw the rectangle at a scale twice as large on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q2/5

1287

QUESTION ID

S37-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A rectangle 8 units by 4 units, in any position or orientation

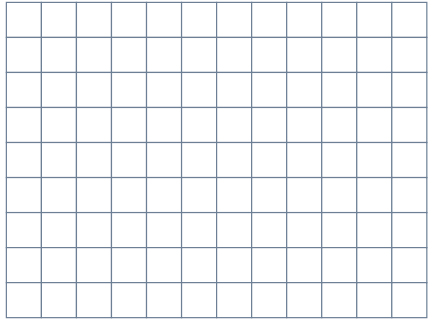
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q2/5

A floor plan shows a room as a rectangle 9 units by 6 units. Its scale is 1 unit : 2 feet.

Redraw the room on the grid at a scale of 1 unit : 3 feet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q3/5

1288

QUESTION ID

S37-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

A rectangle 6 units by 4 units, in any position or orientation

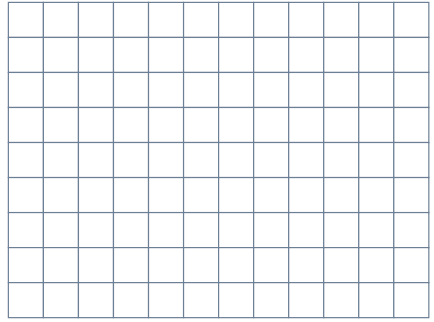
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q3/5

A scale drawing of a ramp is a right triangle whose two sides that form the right angle are 3 units and 4 units. Its scale is 1 unit : 5 m.

Redraw it on the grid at a scale of 1 unit : 2.5 m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q4/5

1289

QUESTION ID

S37-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A right triangle whose two sides that form the right angle are 6 units and 8 units, in any position or orientation

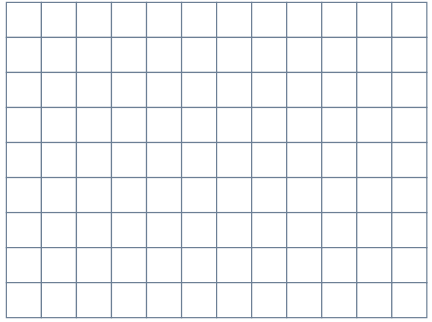
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q4/5

A scale drawing of a patio is a rectangle 5 units by 3 units. Its scale is 1 unit : 6 m.

Redraw the patio on the grid at a scale of 1 unit : 3 m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q5/5

1290

QUESTION ID

S37-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A rectangle 10 units by 6 units, in any position or orientation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q5/5

SET 37 • 7.G.A.1

BACKWARD 1

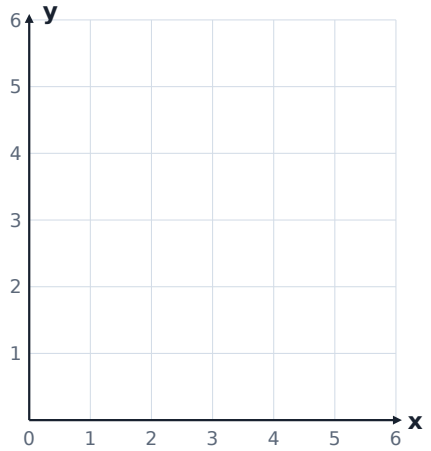
Draw a polygon on the coordinate plane • Grade 6 • Geometry • 6.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Draw the polygon with vertices (1, 1), (5, 1), (5, 4), and (1, 4).

How long is the side from (1, 1) to (5, 1)?

Length: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

1292

QUESTION ID

S37-B1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A rectangle with vertices (1, 1), (5, 1), (5, 4), and (1, 4)

Written: 4 units

GRADING NOTE Grade both: the rectangle drawn with the four given vertices, and the length 4 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

A rectangle has vertices (2, 1), (2, 6), (7, 6), and (7, 1).

How long is the side from (2, 1) to (7, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

1293

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

Which point makes a square with (1, 1), (4, 1), and (4, 4)?

A. (4, 7)

B. (3, 1)

C. (1, 3)

D. (1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

1294

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

True or false?

The points (0, 2) and (0, 7) are 5 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

1295

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

True or false?

The side from (3, 2) to (3, 8) is 5 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

1296

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 6 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 7.G.A.1

BACKWARD 2

Compare the size of a product to the size of one factor • Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a

Without multiplying, is 3 × 12 greater than or less than 12?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q1/5

1298

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Greater than 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q1/5

Without multiplying, is $\frac{1}{2} \times 18$ greater than or less than 18?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q2/5

1299

QUESTION ID

S37-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Less than 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q2/5

Which product is greater than 15?

A. 2×15

B. $\frac{3}{4} \times 15$

C. $\frac{1}{2} \times 15$

D. $\frac{1}{3} \times 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q3/5

1300

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2 × 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3} \times 9$ is less than 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q4/5

1301

QUESTION ID

S37-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q4/5

True or false?

4 × 8 is less than 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q5/5

1302

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 7.G.A.1

FORWARD 1

Describe the effect of a dilation on coordinates • Grade 8 • Geometry • 8.G.A.3

The point (4, −2) is dilated by a scale factor of 3 with the origin as the center.

What are the coordinates of the image?

Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q1/5

1304

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(12, −6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q1/5

A triangle has vertices (2, 2), (6, 2), and (2, 4). It is dilated by a scale factor of $\frac{1}{2}$ with the origin as the center.

What are the vertices of the image?

Vertices: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q2/5

1305

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 1), (3, 1), and (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q2/5

A dilation centered at the origin maps (5, 10) to (2, 4).

What is the scale factor?

A. $\frac{2}{5}$

B. $\frac{5}{2}$

C. -3

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q3/5

1306

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q3/5

True or false?

A dilation by a scale factor of 2 with the origin as the center maps (−3, 1) to (−6, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q4/5

1307

QUESTION ID

S37-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q4/5

True or false?

A dilation by a scale factor of 4 with the origin as the center maps (2, 3) to (6, 7).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q5/5

1308

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It maps (2, 3) to (8, 12).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 7.G.A.1

FORWARD 2

Verify the properties of dilations • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

A segment 5 cm long is dilated by a scale factor of 1.8.

How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q1/5

1310

QUESTION ID

S37-F2-Q1

TYPE

Short answer

CORRECT ANSWER

9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q1**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q1/5

A line passes through the center of a dilation.

What is the image of the line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q2/5

1311

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The same line (it maps to itself).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q2/5

A line that does not pass through the center of a dilation is dilated.

What is its image?

A. The same line

B. A curve

C. A perpendicular line

D. A parallel line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q3/5

1312

QUESTION ID

S37-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A parallel line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q3/5

True or false?

A dilation with a scale factor between 0 and 1 makes segments shorter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q4/5

1313

QUESTION ID

S37-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q4/5

True or false?

A dilation always changes the slope of a line that does not pass through its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q5/5

1314

QUESTION ID

S37-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q5/5

SET 38 • 7.G.A.2 • GEOMETRY

Draw triangles from given conditions

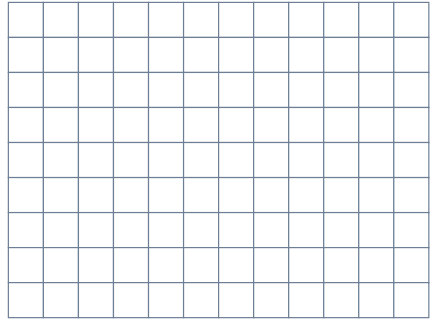
MAIN	Grade 7 main question	**7.G.A.2**
BACKWARD 1	Measure and draw angles with a protractor	**4.MD.C.6**
BACKWARD 2	Classify triangles by their angles	**4.G.A.2**
FORWARD 1	Use the Pythagorean theorem to find an unknown side length	**8.G.B.7**
FORWARD 2	Make formal constructions with a compass and straightedge	**HSG-CO.D.12**

SET 38 • 7.G.A.2

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.2

On the grid, draw a triangle with a right angle. The two sides that form the right angle should be 4 units and 6 units long.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q1/5

1317

QUESTION ID

S38-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Any right triangle whose two sides that form the right angle are 4 units and 6 units long, in any position or orientation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a ruler and a protractor. Draw a triangle with a 5 cm side that has a 50° angle at one end and a 60° angle at the other end.

Measure the third angle.

Third angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q2/5

1318

QUESTION ID

S38-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A triangle with a 5 cm side whose two end angles measure 50° and 60°, in any position or orientation

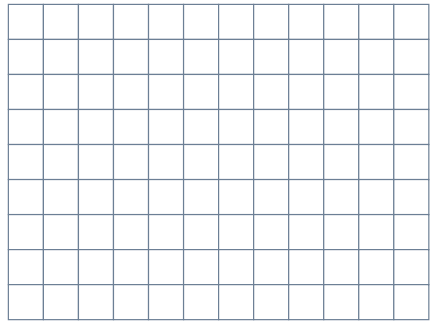
Written: 70°

GRADING NOTE Grade both: the triangle drawn with the given side and angles (within ±2 mm and ±2°), and the third angle 70°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q2/5

On the grid, draw an obtuse triangle that has one side 5 units long along a grid line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q3/5

1319

QUESTION ID

S38-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

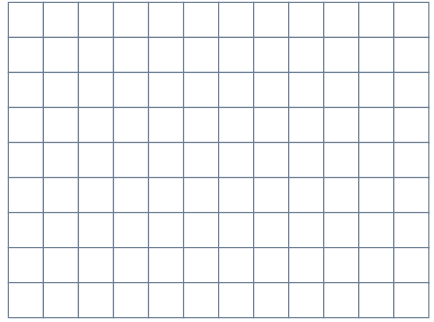
Any triangle with one angle greater than 90° and one side 5 units long along a grid line, in any position or orientation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q3/5

On the grid, draw an isosceles triangle that is not equilateral.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q4/5

1320

QUESTION ID

S38-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

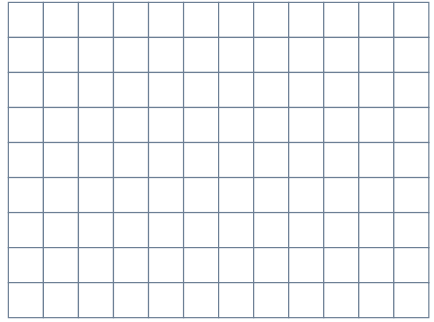
Any triangle with exactly two sides of equal length (the third side has a different length)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q4/5

On the grid, draw a triangle with two sides that are each 5 units long and meet at a right angle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q5/5

1321

QUESTION ID

S38-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A right triangle whose two sides that form the right angle are both 5 units long (an isosceles right triangle), in any position or orientation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q5/5

SET 38 • 7.G.A.2

BACKWARD 1

Measure and draw angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

Use a protractor to draw a 65° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q1/5

1323

QUESTION ID

S38-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

An angle that measures 65° (within ±2°), in any orientation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q1/5

One ray of an angle lines up with 0 on a protractor. The other ray passes through 120.

What is the measure of the angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q2/5

1324

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

120°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q2/5

Which angle is greater than a right angle?

A. 80°

B. 90°

C. 110°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q3/5

1325

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 110°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q3/5

True or false?

A protractor measures angles in degrees.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q4/5

1326

QUESTION ID

S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q4/5

True or false?

A 30° angle is an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q5/5

1327

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • 7.G.A.2

BACKWARD 2

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

A triangle has a 90° angle.

Classify the triangle by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q1/5

1329

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q1/5

A triangle has angles of 100°, 50°, and 30°.

Classify the triangle by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q2/5

1330

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Obtuse triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q2/5

What kind of triangle has three angles that are each less than 90°?

A. Obtuse triangle

B. Acute triangle

C. Straight triangle

D. Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q3/5

1331

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Acute triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q3/5

True or false?

An obtuse triangle has exactly one obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q4/5

1332

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q4/5

True or false?

A right triangle can have two right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q5/5

1333

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 7.G.A.2

FORWARD 1

Use the Pythagorean theorem to find an unknown side length • Grade 8 • Geometry • 8.G.B.7

A right triangle has legs of 6 cm and 8 cm.

How long is the hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q1/5

1335

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q1/5

A right triangle has a hypotenuse of 13 m and one leg of 5 m.

How long is the other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q2/5

1336

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q2/5

A 10-foot ladder leans against a wall. The top of the ladder is 8 feet above the ground.

How far is the bottom of the ladder from the wall?

A. 6 feet

B. 12.8 feet

C. 18 feet

D. 2 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q3/5

1337

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q3/5

True or false?

A right triangle with legs of 9 and 12 has a hypotenuse of 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q4/5

1338

QUESTION ID

S38-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q4/5

True or false?

A right triangle with legs of 1 and 1 has a hypotenuse of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q5/5

1339

QUESTION ID

S38-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The hypotenuse is √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q5/5

SET 38 • 7.G.A.2

FORWARD 2

Make formal constructions with a compass and straightedge • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12

Use a compass and a straightedge to construct the perpendicular bisector of segment AB. Leave your construction arcs showing.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q1/5

1341

QUESTION ID

S38-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Two pairs of arcs with the same radius (more than half of AB), centered at A and at B, meeting above and below AB, and the line through the two meeting points. It crosses AB at its midpoint at a right angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q1/5

Use a compass and a straightedge to construct an equilateral triangle with side AB. Leave your construction arcs showing.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q2/5

1342

QUESTION ID

S38-F2-Q2

TYPE

Drawing / plotting

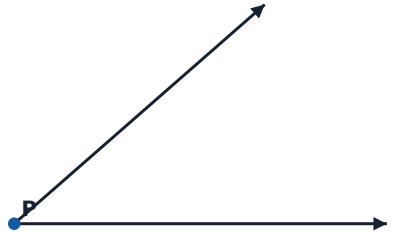
CORRECT ANSWER

Arcs with radius AB centered at A and at B that meet at a point C (above or below AB), with segments AC and BC drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q2/5

Use a compass and a straightedge to bisect the angle at P. Leave your construction arcs showing.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q3/5

1343

QUESTION ID

S38-F2-Q3

TYPE

Drawing / plotting

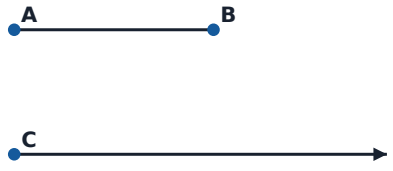
CORRECT ANSWER

An arc centered at P crossing both rays, two arcs with the same radius centered at those crossing points that meet inside the angle, and the ray from P through that meeting point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q3/5

Use a compass and a straightedge to copy segment AB onto the ray, starting at point C. Leave your construction arc showing.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q4/5

1344

QUESTION ID

S38-F2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

An arc centered at C with radius AB (the compass set to the length AB) that crosses the ray at a point D, so CD = AB. D may be labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q4/5

True or false?

In a compass-and-straightedge construction, a protractor is used to measure angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q5/5

1345

QUESTION ID
S38-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only a compass and a straightedge (with no measuring marks) are used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q5/5

SET 39 • 7.G.A.2 • GEOMETRY

Decide whether given conditions make one triangle, more than one triangle, or no triangle

MAIN	Grade 7 main question	**7.G.A.2**
BACKWARD 1	Find an unknown angle when angles are joined	**4.MD.C.7**
BACKWARD 2	Find the perimeter of a polygon from its side lengths	**3.MD.D.8**
FORWARD 1	Use the converse of the Pythagorean theorem to test for a right tr…	**8.G.B.6**
FORWARD 2	Explain triangle congruence criteria with rigid motions	**HSG-CO.B.8**

SET 39 • 7.G.A.2

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.2

Different triangles means triangles that are not congruent; position and orientation do not matter.

How many different triangles have sides of 3 cm, 4 cm, and 9 cm?

A. Exactly one triangle

B. More than one triangle

C. No triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q1/5

1348

QUESTION ID

S39-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. No triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q1/5

Different triangles means triangles that are not congruent; position and orientation do not matter.

A triangle has angles of 30°, 60°, and 90°. No side lengths are given.

How many different triangles fit these conditions? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q2/5

1349

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

More than one. Triangles of different sizes can have the same three angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q2/5

Different triangles means triangles that are not congruent; position and orientation do not matter.

How many different triangles have sides of 5 cm, 6 cm, and 7 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q3/5

1350

QUESTION ID

S39-M-Q3

TYPE

Short answer

CORRECT ANSWER

Exactly one triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q3/5

True or false?

Angles of 80° and 110° cannot both be angles of the same triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q4/5

1351

QUESTION ID

S39-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q4/5

True or false?

Three side lengths that form a triangle can make two triangles that are not congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q5/5

1352

QUESTION ID

S39-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q5/5

SET 39 • 7.G.A.2

BACKWARD 1

Find an unknown angle when angles are joined • Grade 4 • Measurement & Data • 4.MD.C.7

Angle ABC is split into two angles that measure 35° and 50°.

What is the measure of angle ABC?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q1/5

1354

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q1/5

A 90° angle is split into two angles. One measures 28°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q2/5

1355

QUESTION ID

S39-B1-Q2

TYPE

Short answer

CORRECT ANSWER

62°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q2/5

A 120° angle is split into two angles. One measures 45°.

What does the other angle measure?

A. 45°

B. 165°

C. 75°

D. 85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q3/5

1356

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 75°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A 45° angle and a 45° angle joined together make a 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q4/5

1357

QUESTION ID

S39-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q4/5

True or false?

A 120° angle and a 70° angle joined together make a 180° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q5/5

1358

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They make a 190° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 7.G.A.2

BACKWARD 2

Find the perimeter of a polygon from its side lengths • Grade 3 • Measurement & Data • 3.MD.D.8

A triangle has sides of 7 cm, 9 cm, and 12 cm.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q1/5

1360

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

28 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q1/5

A triangle has a perimeter of 30 inches. Two of its sides are 8 inches and 10 inches.

How long is the third side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q2/5

1361

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q2/5

A triangle has sides of 5 m, 5 m, and 6 m.

What is its perimeter?

A. 11 m

B. 21 m

C. 150 m

D. 16 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q3/5

1362

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 16 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q3/5

True or false?

A triangle with sides of 4 feet, 6 feet, and 7 feet has a perimeter of 17 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q4/5

1363

QUESTION ID

S39-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q4/5

True or false?

A triangle with three 9 cm sides has a perimeter of 18 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q5/5

1364

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • 7.G.A.2

FORWARD 1

Use the converse of the Pythagorean theorem to test for a right triangle • Grade 8 • Geometry • 8.G.B.6

Is a triangle with sides of 7, 24, and 25 a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q1/5

1366

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes.

$7^2 + 24^2 = 49 + 576 = 625 = 25^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q1/5

Is a triangle with sides of 6, 7, and 9 a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q2/5

1367

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. $6^2 + 7^2 = 85$, but $9^2 = 81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q2/5

Which side lengths form a right triangle?

A. 8, 15, 17

B. 4, 5, 6

C. 5, 6, 8

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q3/5

1368

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8, 15, 17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q3/5

True or false?

A triangle with sides of 9, 40, and 41 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q4/5

1369

QUESTION ID

S39-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q4/5

True or false?

A triangle with sides of 3, 3, and 4 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q5/5

1370

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 7.G.A.2

FORWARD 2

Explain triangle congruence criteria with rigid motions • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8

Triangles ABC and DEF have AB = DE, BC = EF, and angle B = angle E. A rigid motion moves B onto E and ray BA onto ray ED.

Explain why A lands on D and C lands on F.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q1/5

1372

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A lands on D because AB = DE along the same ray. Because angle B = angle E, ray BC lands on ray EF (after a reflection if needed), and because BC = EF, C lands on F. So the triangles are congruent (SAS).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q1/5

Triangles PQR and STU have PQ = ST, QR = TU, and PR = SU. A rigid motion moves P onto S and Q onto T, with R and U on the same side of ST.

Explain why R must land on U.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q2/5

1373

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

R lands where a circle of radius PR around S meets a circle of radius QR around T on that side of ST. U is that same point (SU = PR and TU = QR), and there is only one such point, so R lands on U (SSS).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q2/5

Triangles ABC and DEF have angle A = angle D, AB = DE, and angle B = angle E. A rigid motion puts A on D and B on E, with C and F on the same side.

Why must C land on F?

A. All triangles with a side of length AB are congruent.

B. A rigid motion stretches the sides until they match.

C. The third angles add to 180°.

D. The rays from D and E at the equal angles cross at only one point, so C and F are the same point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q3/5

1374

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The rays from D and E at the equal angles cross at only one point, so C and F are the same point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q3/5

True or false?

Rigid motions (translations, rotations, and reflections) keep side lengths and angle measures the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q4/5

1375

QUESTION ID

S39-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q4/5

True or false?

Two triangles with all three pairs of angles equal can always be mapped onto each other by a rigid motion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q5/5

1376

QUESTION ID
S39-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Equal angles make the triangles similar; they can be different sizes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q5/5

SET 40 • 7.G.A.3 • GEOMETRY

Describe the cross sections made by slicing right prisms and pyramids

MAIN	Grade 7 main question	**7.G.A.3**
BACKWARD 1	Identify the faces of solids from their nets	**6.G.A.4**
BACKWARD 2	Name shapes by their number of sides and faces	**2.G.A.1**
FORWARD 1	Find the volumes of cylinders, cones, and spheres	**8.G.C.9**
FORWARD 2	Identify cross sections of three-dimensional objects	**HSG-GMD.B.4**

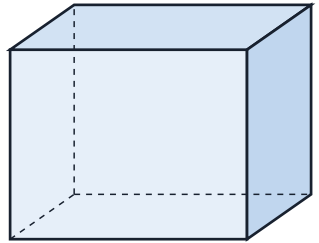
SET 40 • 7.G.A.3

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.3

A right rectangular prism is sliced by a plane parallel to its base.

What is the shape of the cross section?



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q1/5

1379

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle the same size as the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q1/5

A right square pyramid is sliced by a plane parallel to its base, strictly between the base and the top vertex.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q2/5

1380

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

A square smaller than the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q2/5

A cube is sliced by a plane that is parallel to one of its faces.

What is the shape of the cross section?

A. A triangle

B. A pentagon

C. A square

D. A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q3/5

1381

QUESTION ID

S40-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q3/5

True or false?

A right square pyramid is sliced by a plane that passes through its top vertex and the center of its base, perpendicular to the base. The cross section is a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q4/5

1382

QUESTION ID

S40-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q4/5

True or false?

A right rectangular prism sliced by a plane parallel to its base can have a triangle as its cross section.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q5/5

1383

QUESTION ID

S40-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q5/5

SET 40 • 7.G.A.3

BACKWARD 1

Identify the faces of solids from their nets • Grade 6 • Geometry • 6.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

How many faces does a rectangular prism have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q1/5

1385

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q1/5

A net has 1 square and 4 triangles.

What solid does it fold into?

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q2/5

1386

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A square pyramid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q2/5

Which solid has a net of 2 triangles and 3 rectangles?

A. Square pyramid

B. Rectangular prism

C. Triangular prism

D. Cube

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q3/5

1387

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Triangular prism

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q3/5

True or false?

The net of a cube has 6 squares.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q4/5

1388

QUESTION ID

S40-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q4/5

True or false?

A triangular pyramid has 5 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q5/5

1389

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 4 faces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 7.G.A.3

BACKWARD 2

Name shapes by their number of sides and faces • Grade 2 • Geometry • 2.G.A.1

How many sides does a pentagon have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q1/5

1391

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q1/5

A shape has 6 sides and 6 angles.

What is the name of the shape?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q2/5

1392

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q2/5

Which shape has 4 sides?

A. Quadrilateral

B. Hexagon

C. Pentagon

D. Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q3/5

1393

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Quadrilateral

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q3/5

True or false?

A cube has 6 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q4/5

1394

QUESTION ID

S40-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q4/5

True or false?

A triangle has 4 angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q5/5

1395

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • 7.G.A.3

FORWARD 1

Find the volumes of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 2 cm and a height of 9 cm.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q1/5

1397

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

113.04 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q1/5

A cone has a radius of 6 inches and a height of 5 inches.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q2/5

1398

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60π cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q2/5

A sphere has a radius of 3 m.

What is its volume in terms of π?

A. 12π cubic meters

B. 108π cubic meters

C. 27π cubic meters

D. 36π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q3/5

1399

QUESTION ID

S40-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 36π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q3/5

True or false?

A cone and a cylinder have the same base and height. The cone's volume is $\frac{1}{3}$ of the cylinder's volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q4/5

1400

QUESTION ID

S40-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 2 and a height of 5 has a volume of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q5/5

1401

QUESTION ID

S40-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 20π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q5/5

SET 40 • 7.G.A.3

FORWARD 2

Identify cross sections of three-dimensional objects • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A right circular cylinder is sliced by a plane parallel to its base that cuts through its interior.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q1/5

1403

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q1**

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q1/5

A right circular cone is sliced by a plane that contains its axis (the line through the apex and the center of the base).

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q2/5

1404

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A triangle (an isosceles triangle)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q2/5

A sphere is sliced by a plane that passes through its interior.

What is the shape of the cross section?

A. A triangle

B. A square

C. An oval that is not a circle

D. A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q3/5

1405

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q3/5

True or false?

Spinning a rectangle around one of its sides makes a cylinder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q4/5

1406

QUESTION ID

S40-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q4/5

True or false?

A cube can never have a cross section shaped like a hexagon.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q5/5

1407

QUESTION ID

S40-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A slice through the midpoints of six edges makes a hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q5/5

SET 41 • 7.G.B.4 • GEOMETRY

Find the circumference of a circle

MAIN	Grade 7 main question	**7.G.B.4**
BACKWARD 1	Find the perimeter of a rectangle	**4.MD.A.3**
BACKWARD 2	Multiply decimals	**5.NBT.B.7**
FORWARD 1	Find the volume of a cylinder	**8.G.C.9**
FORWARD 2	Find the length of an arc of a circle	**HSG-C.B.5**

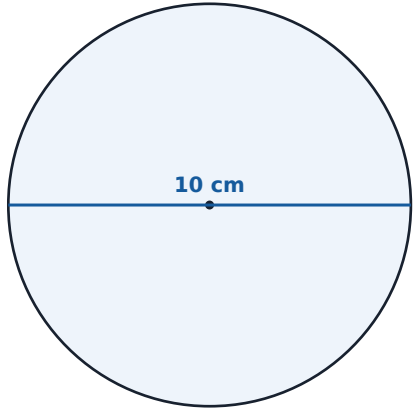
SET 41 • 7.G.B.4

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.4

Find the circumference of the circle. Use 3.14 for π.

Circumference: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q1/5

1410

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

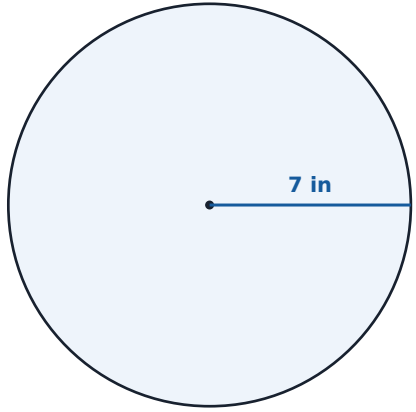
31.4 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q1/5

Find the circumference of the circle. Use $\frac{22}{7}$ for π.



Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q2/5

1411

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

44 in

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q2/5

A circular track has a circumference of 62.8 m.

What is its diameter? Use 3.14 for π.

A. 31.4 m

B. 20 m

C. 10 m

D. 197.2 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q3/5

1412

QUESTION ID

S41-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q3/5

True or false?

A wheel has a radius of 15 inches. In one full turn it rolls about 94.2 inches. (Use 3.14 for π.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q4/5

1413

QUESTION ID

S41-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A circle with a radius of 4 m has a circumference of 4π m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q5/5

1414

QUESTION ID

S41-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The circumference is 8π m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q5/5

SET 41 • 7.G.B.4

BACKWARD 1

Find the perimeter of a rectangle • Grade 4 • Measurement & Data • 4.MD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle is 9 m long and 4 m wide.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q1/5

1416

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

26 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q1/5

A square has 8 cm sides.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q2/5

1417

QUESTION ID

S41-B1-Q2

TYPE

Short answer

CORRECT ANSWER

32 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q2/5

A rectangle is 12 feet long and 5 feet wide.

What is its perimeter?

A. 29 feet

B. 17 feet

C. 60 feet

D. 34 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q3/5

1418

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 34 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q3/5

True or false?

A rectangle 6 inches long and 3 inches wide has a perimeter of 18 inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q4/5

1419

QUESTION ID

S41-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q4/5

True or false?

A square with 10 m sides has a perimeter of 100 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q5/5

1420

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The perimeter is 40 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 7.G.B.4

BACKWARD 2

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

3.14 × 6

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q1/5

1422

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

18.84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q1/5

Multiply.

3.14 × 2.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q2/5

1423

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7.85

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q2/5

Multiply.

3.14 × 12

A. 37.68

B. 36.48

C. 376.8

D. 15.14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q3/5

1424

QUESTION ID

S41-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 37.68

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q3/5

True or false?

$3.14 \times 10 = 31.4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q4/5

1425

QUESTION ID

S41-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q4/5

True or false?

$3.14 \times 4 = 12.16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q5/5

1426

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • 7.G.B.4

FORWARD 1

Find the volume of a cylinder • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 5 feet and a height of 4 feet.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q1/5

1428

QUESTION ID

S41-F1-Q1

TYPE

Short answer

CORRECT ANSWER

100π cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q1/5

A cylinder has a diameter of 6 cm and a height of 10 cm.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q2/5

1429

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

282.6 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q2/5

A cylinder has a volume of 90π cubic meters and a radius of 3 m.

What is its height?

A. 30 m

B. 3 m

C. 10 m

D. 15 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q3/5

1430

QUESTION ID

S41-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q3/5

True or false?

Doubling the height of a cylinder doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q4/5

1431

QUESTION ID

S41-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 1 and a height of 1 has a volume of 2π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q5/5

1432

QUESTION ID

S41-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q5/5

SET 41 • 7.G.B.4

FORWARD 2

Find the length of an arc of a circle • High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5

A circle has a radius of 6 cm.

Find the length of an arc with a central angle of 90°, in terms of π.

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q1/5

1434

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3π cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q1/5

A circle has a circumference of 40 inches.

How long is an arc with a central angle of 45°?

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q2/5

1435

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q2/5

A circle has a radius of 10 m.

What is the length of an arc with a central angle of 60°?

A. $\frac{20}{3}\pi$ m

B. $\frac{5}{3}\pi$ m

C. $\frac{10}{3}\pi$ m

D. 60π m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q3/5

1436

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10/3π m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q3/5

Explain why an arc with a central angle of n° in a circle of radius r has a length of $\frac{n}{360} \times 2\pi r$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q4/5

1437

QUESTION ID

S41-F2-Q4

TYPE

Short answer

CORRECT ANSWER

The arc is n/360 of the whole circle, so its length is n/360 of the circumference, which is 2πr.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q4/5

True or false?

In a circle with a radius of 3, an arc with a central angle of 120° has a length of 3π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q5/5

1438

QUESTION ID
S41-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The arc length is 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q5/5

SET 42 • 7.G.B.4 • GEOMETRY

Find the area of a circle

MAIN	Grade 7 main question	**7.G.B.4**
BACKWARD 1	Evaluate powers with whole-number exponents	**6.EE.A.1**
BACKWARD 2	Evaluate a formula	**6.EE.A.2.c**
FORWARD 1	Find the volume of a cone	**8.G.C.9**
FORWARD 2	Find the area of a sector of a circle	**HSG-C.B.5**

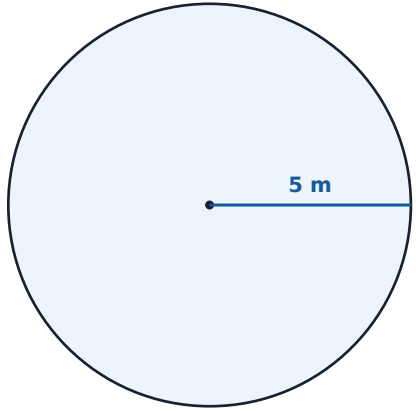
SET 42 • 7.G.B.4

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.4

Find the area of the circle.
Use 3.14 for π.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q1/5

1441

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

78.5 square meters

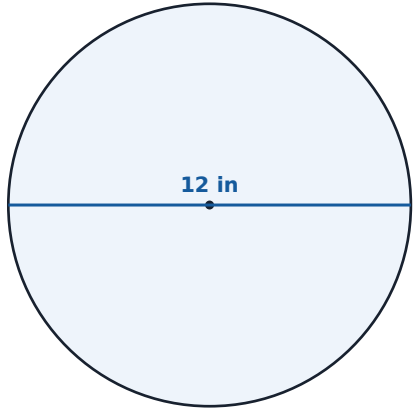
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q1/5

Find the area of the circle in terms of π.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q2/5

1442

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

36π square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q2/5

A circular rug has a radius of 3 feet.

What is its area? Use 3.14 for π.

A. 113.04 square feet

B. 28.26 square feet

C. 9.42 square feet

D. 18.84 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q3/5

1443

QUESTION ID

S42-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 28.26 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q3/5

True or false?

A circle with a radius of 10 cm has an area of 100π square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q4/5

1444

QUESTION ID

S42-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q4/5

True or false?

A circle with a diameter of 8 m has an area of 64π square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q5/5

1445

QUESTION ID

S42-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The radius is 4 m, so the area is 16π square meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q5/5

SET 42 • 7.G.B.4

BACKWARD 1

Evaluate powers with whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

7^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q1/5

1447

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

1.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q2/5

1448

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q2/5

Find the value.

4^3

A. 7

B. 64

C. 12

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q3/5

1449

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q3/5

True or false?

$9^2 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q4/5

1450

QUESTION ID

S42-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q4/5

True or false?

$6^2 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q5/5

1451

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 7.G.B.4

BACKWARD 2

Evaluate a formula • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use $A = s^2$.

Find A when s = 11.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q1/5

1453

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

121

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q1/5

Use $A = 3.14r^2$.

Find A when r = 2.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q2/5

1454

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12.56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q2/5

Use $A = \frac{1}{2}bh$.

Find A when b = 8 and h = 5.

A. 13

B. 20

C. 40

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q3/5

1455

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q3/5

True or false?

Use $V = s^3$. When $s = 2$, $V = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q4/5

1456

QUESTION ID

S42-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q4/5

True or false?

Use A = lw. When l = 3.5 and w = 2, A = 5.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q5/5

1457

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q5/5

SET 42 • 7.G.B.4

FORWARD 1

Find the volume of a cone • Grade 8 • Geometry • 8.G.C.9

A cone has a radius of 3 cm and a height of 8 cm.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q1/5

1459

QUESTION ID

S42-F1-Q1

TYPE

Short answer

CORRECT ANSWER

24π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q1/5

A cone has a radius of 6 feet and a height of 7 feet.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q2/5

1460

QUESTION ID

S42-F1-Q2

TYPE

Short answer

CORRECT ANSWER

263.76 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q2/5

A cone has a base area of 30 square inches and a height of 9 inches.

What is its volume?

A. 270 cubic inches

B. 90 cubic inches

C. 30 cubic inches

D. 39 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q3/5

1461

QUESTION ID

S42-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 90 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q3/5

True or false?

A cone with a radius of 3 and a height of 3 has a volume of 9π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q4/5

1462

QUESTION ID

S42-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q4/5

True or false?

The volume of a cone is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q5/5

1463

QUESTION ID

S42-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The volume of a cone is $(1/3)\pi r^2 h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q5/5

SET 42 • 7.G.B.4

FORWARD 2

Find the area of a sector of a circle • High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5

A circle has a radius of 4 m.

Find the area of a sector with a central angle of 90°, in terms of π.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q1/5

1465

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q1/5

A pizza with a radius of 8 inches is cut into 8 equal slices.

What is the area of one slice, in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q2/5

1466

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

8π square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q2/5

A circle has a radius of 6 cm.

What is the area of a sector with a central angle of 120°?

A. 12π square centimeters **B.** 24π square centimeters

C. 4π square centimeters **D.** 36π square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q3/5

1467

QUESTION ID

S42-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12π square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q3/5

Explain why a sector with a central angle of n° in a circle of radius r has an area of $\frac{n}{360} \times \pi r^2$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q4/5

1468

QUESTION ID

S42-F2-Q4

TYPE

Short answer

CORRECT ANSWER

The sector is n/360 of the whole circle, so its area is n/360 of the circle's area, which is πr².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q4/5

True or false?

In a circle with a radius of 6, a sector with a central angle of 60° has an area of 12π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q5/5

1469

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The area is 6π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q5/5

SET 43 • 7.G.B.4 • GEOMETRY

Explain the relationship between the circumference and the area of a circle

MAIN	Grade 7 main question	**7.G.B.4**
BACKWARD 1	Find the area of a parallelogram	**6.G.A.1**
BACKWARD 2	Identify equivalent expressions	**6.EE.A.4**
FORWARD 1	Use square roots to solve equations of the form $x^2 = p$	**8.EE.A.2**
FORWARD 2	Give an informal argument for circle and volume formulas	**HSG-GMD.A.1**

SET 43 • 7.G.B.4

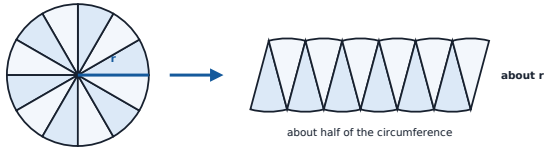
MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The circle with radius r is cut into equal wedges and rearranged into a shape close to a parallelogram. Its base is half the circumference.

Write the base in terms of r.



Base: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q1/5

1472

QUESTION ID

S43-M-Q1

TYPE

Short answer

CORRECT ANSWER

πr

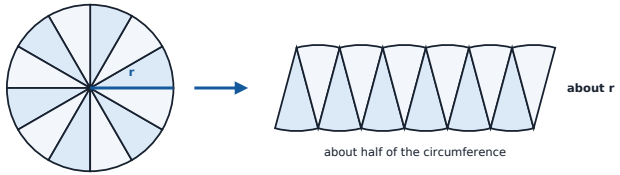
GRADING NOTE Half of 2πr.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q1/5

The circle with radius r is cut into equal wedges and rearranged as shown.

Explain how this shows that the area of a circle is πr^2.



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q2/5

1473

QUESTION ID

S43-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

πr^2

GRADING NOTE REQUIRED WORK: A complete argument: the rearranged shape is close to a parallelogram (a rectangle with more wedges); its base is half the circumference, $\frac{1}{2} \times 2\pi r = \pi r$; its height is the radius r; so its area is $\pi r \times r = \pi r^2$, and the circle has the same area because it is made of the same wedges. Any equivalent wording is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q2/5

A circle rearranged into a near-parallelogram has a base of πr and a height of r.

What is the area?

A. πr^2

B. $2\pi r^2$

C. πr

D. $2\pi r$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q3/5

1474

QUESTION ID

S43-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. πr^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q3/5

True or false?

A circle has circumference C and radius r. Its area equals $\frac{1}{2} \times C \times r$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q4/5

1475

QUESTION ID

S43-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q4/5

True or false?

A circle with a circumference of 10π has an area of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q5/5

1476

QUESTION ID

S43-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its radius is 5, so its area is 25π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q5/5

SET 43 • 7.G.B.4

BACKWARD 1

Find the area of a parallelogram • Grade 6 • Geometry • 6.G.A.1

A parallelogram has a base of 9 cm and a height of 4 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

1478

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

A parallelogram has a base of 12 m and a height of 6.5 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

1479

QUESTION ID

S43-B1-Q2

TYPE

Short answer

CORRECT ANSWER

78 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

A parallelogram is cut along a segment from one top corner straight down to the base, at a right angle to the base. The triangle that is cut off is slid to the other end, so its slanted side lines up with the other slanted side.

What shape is made?

A. A pentagon

B. A trapezoid that is not a rectangle

C. A circle

D. A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

1480

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

A parallelogram and a rectangle with the same base and the same height have the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

1481

QUESTION ID

S43-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

A parallelogram has a base of 10 and slanted sides of 6. Its area is 60.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

1482

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The area uses the height, not the slanted side, so it is less than 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 7.G.B.4

BACKWARD 2

Identify equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Is $\frac{1}{2} \times 2 \times r$ equivalent to r? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q1/5

1484

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 1/2 × 2 = 1, and 1 × r = r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q1/5

Which expression is equivalent to 3 × r × r?

A. 6r

B. $3 + r^2$

C. $(3r)^2$

D. $3r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q2/5

1485

QUESTION ID

S43-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $3r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q2/5

Write an expression equivalent to x · x · x using an exponent.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q3/5

1486

QUESTION ID

S43-B2-Q3

TYPE

Short answer

CORRECT ANSWER

x^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q3/5

True or false?

2(4y) and 8y are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q4/5

1487

QUESTION ID

S43-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q4/5

True or false?

a^2 and $2a$ are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q5/5

1488

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • 7.G.B.4

FORWARD 1

Use square roots to solve equations of the form $x^2 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x. Give all solutions.

$x^2 = 64$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q1/5

1490

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 8 and x = −8

GRADING NOTE Both solutions are required; x = ±8 and set notation are accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q1/5

A circle has an area of 49π square centimeters.

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q2/5

1491

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q2/5

Solve for x.

$x^3 = 27$

A. x = 3

B. x = 9

C. x = 6

D. x = 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q3/5

1492

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q3/5

True or false?

$\sqrt{144} = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q4/5

1493

QUESTION ID

S43-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q4/5

True or false?

√50 is a whole number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q5/5

1494

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q5/5

SET 43 • 7.G.B.4

FORWARD 2

Give an informal argument for circle and volume formulas • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Use the wedge argument to explain why the area of a circle is πr^2.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q1/5

1496

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The wedges form a shape close to a parallelogram with base πr (half the circumference) and height r, so the area is $\pi r \times r = \pi r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q1/5

Two stacks of 20 identical coins have the same height. One stack is straight and one is slanted.

Do they have the same volume? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q2/5

1497

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. At every height the cross sections have the same area (Cavalieri's principle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q2/5

The circumference of a circle is $C = \pi d$.

What is the ratio of C to d?

A. 2π

B. π

C. 2

D. r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q3/5

1498

QUESTION ID

S43-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q3/5

True or false?

As a circle is cut into more and more wedges, the rearranged shape gets closer to a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q4/5

1499

QUESTION ID

S43-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q4/5

True or false?

Cavalieri's principle says that any two solids with the same height have the same volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q5/5

1500

QUESTION ID
S43-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The cross sections must also have equal areas at every height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q5/5

SET 44 • 7.G.B.5 • GEOMETRY

Write and solve equations for supplementary and complementary angles

MAIN	Grade 7 main question	**7.G.B.5**
BACKWARD 1	Find an unknown part of a right angle or a straight angle	**4.MD.C.7**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Use the angle sum and exterior angles of triangles	**8.G.A.5**
FORWARD 2	Prove angle relationships formed by parallel lines and a transvers…	**HSG-CO.C.9**

SET 44 • 7.G.B.5

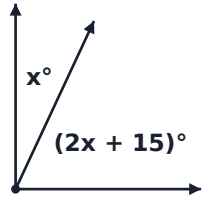
MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, angle ABC is a right angle. A ray splits it into angles of x° and (2x + 15)°.

Write and solve an equation for x.



Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q1/5

1503

QUESTION ID

S44-M-Q1

TYPE

Short answer

CORRECT ANSWER

The equation
x + (2x + 15) = 90
and the answer
x = 25

GRADING NOTE Both parts are required: a valid equation and the correct solution. Any valid equation is accepted, but the solution must also be correct.

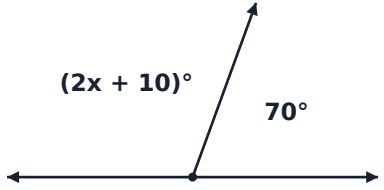
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, angles of $(2x + 10)°$ and $70°$ form a straight line.

Write and solve an equation for x.



Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q2/5

1504

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation
(2x + 10) + 70 = 180
and the answer
x = 50

GRADING NOTE Both parts are required: a valid equation and the correct solution. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q2/5

Two complementary angles measure x° and (x + 20)°.

What is the value of x?

A. 45

B. 35

C. 80

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q3/5

1505

QUESTION ID

S44-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

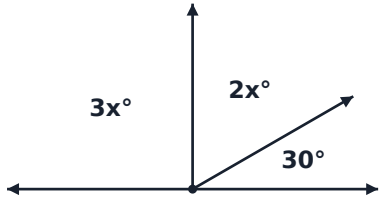
Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q3/5

True or false?

In the figure, three angles form a straight line. Then x = 30.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q4/5

1506

QUESTION ID

S44-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

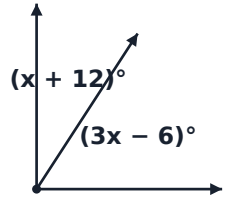
GRADING NOTE 3x + 2x + 30 = 180, so x = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q4/5

True or false?

In the figure, the two angles form a right angle. Then x = 18.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q5/5

1507

GRADING NOTE (3x − 6) + (x + 12) = 90 gives x = 21.

QUESTION ID

S44-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q5/5

SET 44 • 7.G.B.5

BACKWARD 1

Find an unknown part of a right angle or a straight angle • Grade 4 • Measurement & Data • 4.MD.C.7

A straight angle (180°) is split into two angles. One measures 115°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q1/5

1509

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q1/5

A right angle is split into two angles. One measures 34°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q2/5

1510

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

56°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q2/5

A right angle is split into three angles. Two of them measure 20° and 30°.

What does the third angle measure?

A. 50°

B. 60°

C. 40°

D. 10°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q3/5

1511

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q3/5

True or false?

Angles of 72° and 18° together make a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q4/5

1512

QUESTION ID

S44-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q4/5

True or false?

Angles of 95° and 95° together make a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q5/5

1513

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 7.G.B.5

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

x + 48 = 90

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q1/5

1515

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q1/5

Solve for x.

3x = 180

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q2/5

1516

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q2/5

Solve for y.

125 + y = 180

A. y = 55

B. y = 45

C. y = 65

D. y = 305

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q3/5

1517

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. y = 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q3/5

True or false?

a = 27 is the solution of a + 63 = 90.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q4/5

1518

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q4/5

True or false?

b = 80 is the solution of 2b = 180.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q5/5

1519

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • 7.G.B.5

FORWARD 1

Use the angle sum and exterior angles of triangles • Grade 8 • Geometry • 8.G.A.5

Two angles of a triangle measure 48° and 75°.

What does the third angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q1/5

1521

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

57°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q1/5

An exterior angle of a triangle measures 130°. One of the two remote interior angles measures 60°.

What does the other remote interior angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q2/5

1522

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q2/5

The angles of a triangle measure x°, 2x°, and 3x°.

What is the value of x?

A. 36

B. 30

C. 45

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q3/5

1523

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q3/5

True or false?

The interior angles of a triangle add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q4/5

1524

QUESTION ID

S44-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q4/5

True or false?

An exterior angle of a triangle equals the sum of all three interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q5/5

1525

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

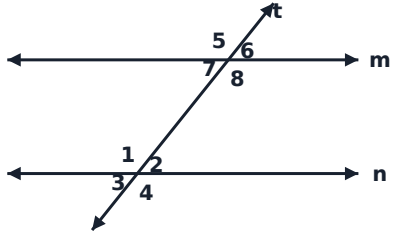
Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • 7.G.B.5

FORWARD 2

Prove angle relationships formed by parallel lines and a transversal • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

In the figure, lines m and n are parallel. Prove that angle 2 = angle 7. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q1/5

1527

QUESTION ID

S44-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 = angle 7

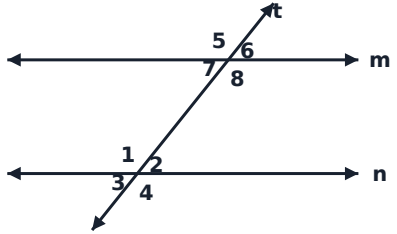
GRADING NOTE REQUIRED WORK: A valid chain of reasons. Example: angle 2 = angle 6 (corresponding angles, because m and n are parallel); angle 6 = angle 7 (vertical angles); so angle 2 = angle 7 (substitution). Other valid chains are accepted. The conclusion without reasons is not full credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Prove that angle 2 + angle 8 = 180°. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q2/5

1528

QUESTION ID

S44-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 + angle 8 = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Example: angle 6 + angle 8 = 180° (they form a linear pair); angle 2 = angle 6 (corresponding angles); so angle 2 + angle 8 = 180° (substitution). Other valid chains are accepted. The conclusion without reasons is not full credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q2/5

In the figure, lines m and n are parallel. A proof says:

Step 1: angle 1 = angle 5 (corresponding angles)

Step 2: angle 5 = angle 8 (?)

Step 3: angle 1 = angle 8

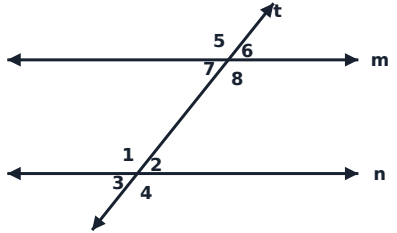
What is the reason for Step 2?

A. Linear pairs are supplementary.

B. Corresponding angles are congruent.

C. Vertical angles are congruent.

D. Alternate exterior angles are congruent.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q3/5

1529

QUESTION ID

S44-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Vertical angles are congruent.

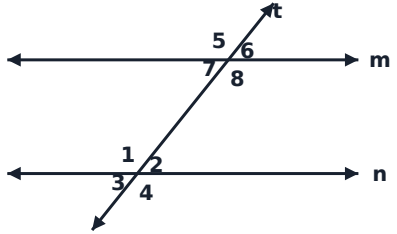
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel and angle 2 = 52°.

Find angle 8 and give a reason for each step.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q4/5

1530

QUESTION ID

S44-F2-Q4

TYPE

Short answer

CORRECT ANSWER

128°: angle 6 = angle 2 = 52° (corresponding angles), and angle 8 = 180° − 52° = 128° (angles 6 and 8 form a linear pair).

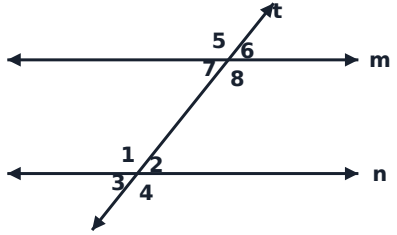
GRADING NOTE Both parts are required: 128° and a valid chain of reasons. A number without reasons is not full credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q4/5

True or false?

In the figure, lines m and n are parallel. "Angle 2 = angle 7 because they look the same size" is a valid proof.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q5/5

1531

QUESTION ID
S44-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A proof needs reasons, such as corresponding angles and vertical angles, not appearance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q5/5

SET 45 • 7.G.B.5 • GEOMETRY

Use vertical and adjacent angles to find unknown angles

MAIN	Grade 7 main question	**7.G.B.5**
BACKWARD 1	Identify lines, rays, segments, and angles	**4.G.A.1**
BACKWARD 2	Understand angle measure as a fraction of a full turn	**4.MD.C.5.a**
FORWARD 1	Find angles formed by parallel lines cut by a transversal	**8.G.A.5**
FORWARD 2	Prove that vertical angles are congruent	**HSG-CO.C.9**

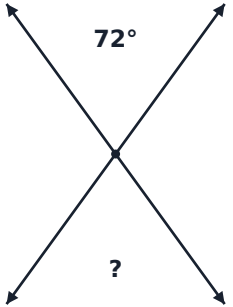
SET 45 • 7.G.B.5

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.5

Two lines intersect. One angle measures 72°.

What does the angle marked with a question mark measure?



Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q1/5

1534

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

72°

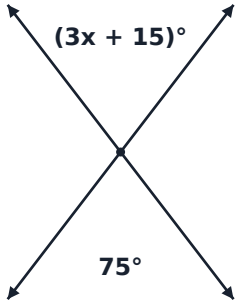
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two lines intersect. The angles shown are vertical angles.

Write and solve an equation for x.



Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q2/5

1535

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 3x + 15 = 75 and the answer x = 20

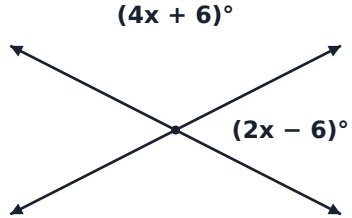
GRADING NOTE Both parts are required: a valid equation and the correct solution. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q2/5

Two lines intersect. The adjacent angles shown form a straight line.

What is the value of x?



A. 15 **B.** 25

C. 30 **D.** 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q3/5

1536

QUESTION ID

S45-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

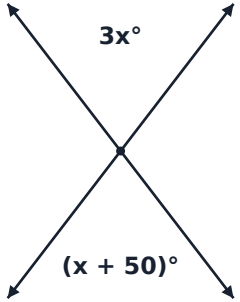
C. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q3/5

True or false?

Two lines intersect. The angles shown are vertical angles. Each of them measures 75°.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q4/5

1537

GRADING NOTE 3x = x + 50, so x = 25 and each angle is 75°.

QUESTION ID

S45-M-Q4

TYPE

True or false

CORRECT ANSWER

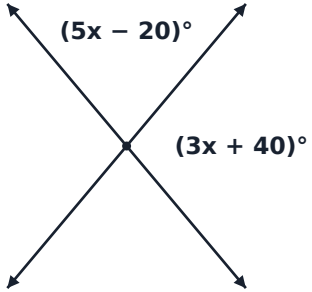
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q4/5

True or false?

Two lines intersect. The adjacent angles shown form a straight line. Then x = 25.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q5/5

1538

GRADING NOTE (5x − 20) + (3x + 40) = 180 gives x = 20.

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q5/5

SET 45 • 7.G.B.5

BACKWARD 1

Identify lines, rays, segments, and angles • Grade 4 • Geometry • 4.G.A.1

How many endpoints does a ray have?

Endpoints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q1/5

1540

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q1/5

What figure is formed by two rays that share an endpoint?

Figure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q2/5

1541

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

An angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q2/5

Which figure goes on forever in both directions?

A. Ray

B. Segment

C. Line

D. Angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q3/5

1542

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q3/5

True or false?

Two lines that cross meet at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q4/5

1543

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q4/5

True or false?

A line segment has no endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q5/5

1544

QUESTION ID

S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 7.G.B.5

BACKWARD 2

Understand angle measure as a fraction of a full turn • Grade 4 • Measurement & Data • 4.MD.C.5.a

How many degrees are in one full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q1/5

1546

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q1/5

An angle turns through $\frac{1}{4}$ of a circle.

What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q2/5

1547

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q2/5

An angle turns through $\frac{1}{2}$ of a circle.

What is its measure?

A. 90°

B. 45°

C. 360°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q3/5

1548

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q3/5

True or false?

An angle that turns through $\frac{1}{360}$ of a circle measures 1°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q4/5

1549

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q4/5

True or false?

An angle that turns through $\frac{1}{3}$ of a circle measures 100°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q5/5

1550

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It measures 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 7.G.B.5

FORWARD 1

Find angles formed by parallel lines cut by a transversal • Grade 8 • Geometry • 8.G.A.5

Two parallel lines are cut by a transversal. One angle measures 110°.

What does the corresponding angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q1/5

1552

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

110°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q1/5

Two parallel lines are cut by a transversal. Alternate interior angles measure (4x − 10)° and 70°.

Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q2/5

1553

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q2/5

Two parallel lines are cut by a transversal. One same-side interior angle measures 75°.

What does the other measure?

A. 15°

B. 105°

C. 75°

D. 285°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q3/5

1554

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 105°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q3/5

True or false?

When parallel lines are cut by a transversal, alternate exterior angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q4/5

1555

QUESTION ID

S45-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q4/5

True or false?

When parallel lines are cut by a transversal, corresponding angles add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q5/5

1556

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 7.G.B.5

FORWARD 2

Prove that vertical angles are congruent • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

Angles 1 and 2 form a linear pair. Angles 2 and 3 form a linear pair.

Explain why angle 1 and angle 3 have the same measure.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q1/5

1558

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

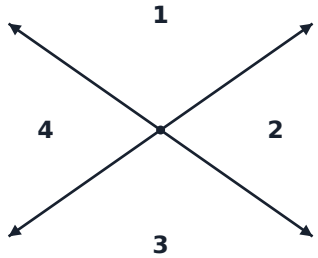
Each one equals 180° minus the measure of angle 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two lines intersect, forming angles 1, 2, 3, and 4 in order around the point. Prove that angle 2 = angle 4. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q2/5

1559

QUESTION ID

S45-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 = angle 4

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Example: angle 1 + angle 2 = 180° and angle 1 + angle 4 = 180° (linear pairs); so angle 2 = 180° − angle 1 = angle 4 (subtraction property of equality). Other valid chains are accepted. The conclusion without reasons is not full credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q2/5

A proof that vertical angles are congruent says:

Step 1: angle 1 + angle 2 = 180°

Step 2: angle 2 + angle 3 = 180°

Step 3: angle 1 + angle 2 = angle 2 + angle 3

Step 4: angle 1 = angle 3

What is the reason for Step 4?

A. Corresponding angles are congruent

B. Vertical angles are congruent

C. Definition of a right angle

D. Subtraction property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q3/5

1560

QUESTION ID

S45-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Subtraction property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two lines intersect, and one of the four angles is 90°.

Prove that all four angles are 90°.

Proof: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q4/5

1561

QUESTION ID

S45-F2-Q4

TYPE

Short answer

CORRECT ANSWER

Each angle next to the 90° angle forms a linear pair with it, so it measures 180° − 90° = 90°. The angle opposite the 90° angle is vertical to it, so it also measures 90°.

GRADING NOTE Full credit needs a reason for each angle (linear pairs and vertical angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q4/5

True or false?

"Angles 1 and 3 are vertical angles, and they look equal, so angle 1 = angle 3" is a valid proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q5/5

1562

QUESTION ID

S45-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A proof needs reasons such as linear pairs, not how the angles look.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q5/5

SET 46 • 7.G.B.6 • GEOMETRY

Find the area of composite figures made of triangles and quadrilaterals

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Find the area of a triangle	**6.G.A.1**
BACKWARD 2	Find areas of rectilinear figures by breaking them into rectangles	**3.MD.C.7.d**
FORWARD 1	Find the volume of a solid made of cylinders, cones, and spheres	**8.G.C.9**
FORWARD 2	Use geometric shapes to model real objects	**HSG-MG.A.1**

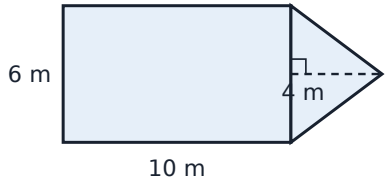
SET 46 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

The figure is a 10 m by 6 m rectangle and a triangle with a base of 6 m (the rectangle's side) and a height of 4 m. The parts do not overlap.

What is the total area?



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q1/5

1565

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

72 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q1/5

An L-shaped floor is made of two rectangles that do not overlap. One is 8 feet by 5 feet and the other is 4 feet by 3 feet.

What is the area of the floor?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q2/5

1566

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

52 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q2/5

A figure is a 10 cm by 4 cm rectangle with a right triangle attached to one 4 cm side. The legs of the triangle are 4 cm and 3 cm, and the parts do not overlap.

What is the total area?

A. 52 square centimeters

B. 34 square centimeters

C. 46 square centimeters

D. 40 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q3/5

1567

QUESTION ID

S46-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 46 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q3/5

True or false?

A 4 cm by 4 cm square is cut from one corner of a 12 cm by 8 cm rectangle. The area that is left is 80 square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q4/5

1568

QUESTION ID

S46-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q4/5

True or false?

A 9 m by 4 m rectangle and a right triangle with legs of 4 m and 3 m are joined without overlapping. Their total area is 48 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q5/5

1569

QUESTION ID

S46-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The total area is 36 + 6 = 42 square meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q5/5

SET 46 • 7.G.B.6

BACKWARD 1

Find the area of a triangle • Grade 6 • Geometry • 6.G.A.1

A triangle has a base of 14 cm and a height of 5 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q1/5

1571

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

35 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q1/5

A right triangle has legs of 9 m and 6 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q2/5

1572

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q2/5

A triangle has a base of 7 feet and a height of 10 feet.

What is its area?

A. 17 square feet

B. 35 square feet

C. 140 square feet

D. 70 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q3/5

1573

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 35 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q3/5

True or false?

A triangle with a base of 8 and a height of 3 has an area of 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q4/5

1574

QUESTION ID

S46-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q4/5

True or false?

A triangle with a base of 6 and a height of 6 has an area of 36.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q5/5

1575

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 7.G.B.6

BACKWARD 2

Find areas of rectilinear figures by breaking them into rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.d

A shape is made of a 5-by-3 rectangle and a 2-by-2 square that do not overlap.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q1/5

1577

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

19 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q1/5

A shape is made of a 6-by-2 rectangle and a 3-by-4 rectangle that do not overlap.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q2/5

1578

QUESTION ID

S46-B2-Q2

TYPE

Short answer

CORRECT ANSWER

24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q2/5

A shape is made of a 4-by-4 square and a 4-by-1 rectangle that do not overlap.

What is its area?

A. 16 square units

B. 20 square units

C. 13 square units

D. 25 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q3/5

1579

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q3/5

True or false?

A 3-by-5 rectangle and a 2-by-5 rectangle that do not overlap have a total area of 25 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q4/5

1580

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q4/5

True or false?

A 7-by-2 rectangle and a 1-by-3 rectangle that do not overlap have a total area of 13 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q5/5

1581

QUESTION ID

S46-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The total area is 17 square units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 7.G.B.6

FORWARD 1

Find the volume of a solid made of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

ANSWER KEY Reduced copy of the question page, with its answer

A cone sits on top of a cylinder. Both have a radius of 3. The cylinder is 10 tall and the cone is 4 tall.

Find the total volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q1/5

1583

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

102π

GRADING NOTE 90π + 12π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q1/5

A hemisphere (half of a sphere) has a radius of 6 cm.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q2/5

1584

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

144π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q2/5

A capsule is a cylinder with radius 2 and height 5 with half of a sphere of radius 2 on each end.

What is its volume?

A. $\frac{52}{3}\pi$

B. 20π

C. $\frac{92}{3}\pi$

D. $\frac{32}{3}\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q3/5

1585

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 92/3π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q3/5

True or false?

A sphere with a radius of 3 has a volume of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q4/5

1586

QUESTION ID

S46-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q4/5

True or false?

A cone with a radius of 3 and a height of 4 has a volume of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q5/5

1587

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 12π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 7.G.B.6

FORWARD 2

Use geometric shapes to model real objects • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1

A tree trunk is modeled as a cylinder with a diameter of 2 feet and a height of 30 feet.

Estimate its volume in terms of π. Then give one reason the model might overestimate or underestimate the real volume.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q1/5

1589

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

30π cubic feet; for example, a real trunk gets narrower toward the top, so the model may overestimate.

GRADING NOTE Grade both: 30π cubic feet, and any reasonable way the real trunk differs from a cylinder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q1/5

A round tabletop is modeled as a circle with a circumference of 9.42 feet.

What is its radius? Use 3.14 for π.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q2/5

1590

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.5 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q2/5

A pile of sand is shaped roughly like a cone that is 6 m across at the bottom and 2 m tall.

Which is the best estimate of its volume?

A. About 18π cubic meters **B.** About 6π cubic meters

C. About 72π cubic meters **D.** About 24π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q3/5

1591

QUESTION ID

S46-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. About 6π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q3/5

True or false?

A rolled sleeping bag 50 cm long and 20 cm across is modeled as a cylinder. Its volume is about 5,000π cubic centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q4/5

1592

QUESTION ID

S46-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q4/5

True or false?

To estimate how much cheese covers a pizza, the pizza is best modeled as a sphere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q5/5

1593

QUESTION ID

S46-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A thin cylinder (or a circle, for the area) is the better model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q5/5

SET 47 • 7.G.B.6 • GEOMETRY

Find the volume of right prisms

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Find the volume of a rectangular prism with fractional edges	**6.G.A.2**
BACKWARD 2	Find volume with $V = l \times w \times h$	**5.MD.C.5.b**
FORWARD 1	Compare the volumes of cylinders and prisms	**8.G.C.9**
FORWARD 2	Use volume formulas to solve problems	**HSG-GMD.A.3**

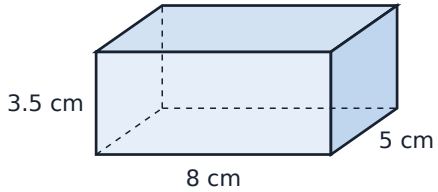
SET 47 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

Find the volume of the right rectangular prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q1/5

1596

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

140 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q1/5

A right triangular prism has triangular bases with a base of 6 m and a height of 4 m. The prism is 10 m long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q2/5

1597

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

120 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q2/5

The base of a right prism has an area of 24 square inches. The prism is 7.5 inches tall.

What is its volume?

A. 90 cubic inches

B. 180 cubic inches

C. 360 cubic inches

D. 31.5 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q3/5

1598

QUESTION ID

S47-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 180 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q3/5

True or false?

A cube with $\frac{1}{2}$-foot edges has a volume of $\frac{1}{8}$ cubic foot.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q4/5

1599

QUESTION ID

S47-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Doubling the height of a right prism doubles the area of its base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q5/5

1600

QUESTION ID

S47-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It doubles the volume; the base does not change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q5/5